

Bladder, prostate and urethra

URINARY BLADDER

The urinary bladder is a reservoir. Its size, shape, position and relations all vary, according to its content and the state of the neighbouring viscera. When the bladder is empty, it lies entirely in the lesser pelvis, but as it distends, it expands anterosuperiorly into the abdominal cavity (Video 75.1). An empty bladder is somewhat tetrahedral and has a base (fundus), neck, apex, and a superior (dome) and two inferolateral surfaces.



RELATIONS

The base of the bladder is triangular and located posteroinferiorly. In females, it is closely related to the anterior vaginal wall (Fig. 75.1); in males, it is related to the rectum, although it is separated from it superiorly by the rectovesical pouch, and inferiorly by the seminal vesicle and vas (ductus) deferens on each side, and by Denonvilliers' fascia (Fig. 75.2). The neck, which is most fixed, lies most inferiorly, 3–4 cm behind the lower part of the pubic symphysis and just above the plane of the inferior aperture of the lesser pelvis. The bladder neck is, essentially, the internal urethral orifice, which lies in a constant position that is independent of the varying positions of the bladder and rectum. In males, the neck rests on, and is in direct continuity with, the base of the prostate; in females, it is related to the pelvic fascia that surrounds the upper urethra. In both sexes, the apex of the bladder faces towards the upper part of the pubic symphysis. The median

umbilical ligament (urachus) ascends behind the anterior abdominal wall from the apex to the umbilicus, covered by peritoneum to form the median umbilical fold (Fig. 75.3, see Fig. 72.19) (see below and p. 1077).

The anterior surface of the bladder is separated from the transversalis fascia by adipose tissue in the potential retropubic space (of Retzius) (Fig. 75.4). This is more adherent to the bladder than to the anterior surface of the prostate, which aids reliable identification of the region of the bladder neck surgically. In males, each inferolateral surface is related anteriorly to the pubis and puboprostatic ligaments. In females, the relations are similar, except that the pubovesical ligaments replace the puboprostatic ligaments. The inferolateral surfaces are not covered by peritoneum. The triangular superior surface is bounded by lateral borders from the apex to the ureteric entrances, and by a posterior border that joins them. In males, the superior surface is completely covered by peritoneum, which extends slightly on to the base and continues posteriorly into the rectovesical pouch and anteriorly into the median umbilical fold; it is in contact with the sigmoid colon and the terminal coils of the ileum (see Fig. 75.2). In females, the superior surface is largely covered by peritoneum, which is reflected posteriorly on to the uterus at the level of the internal os (the junction of the uterine body and cervix), to form the vesicouterine pouch. The posterior part of the superior surface, devoid of peritoneum, is separated from the supravaginal cervix by fibroareolar tissue.

These relationships are important in managing bladder trauma. Extraperitoneal injuries can often be managed conservatively because urine is contained, whereas intraperitoneal injuries usually require surgical repair.

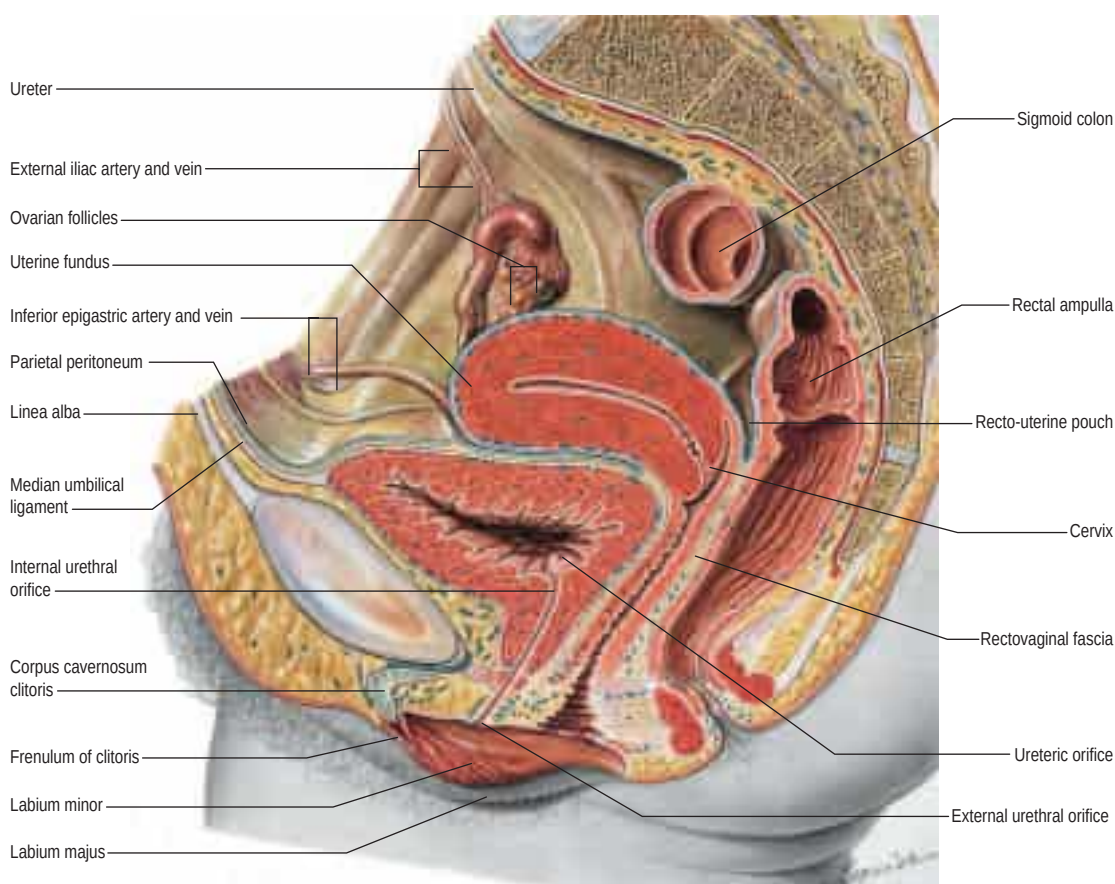


Fig. 75.1 The relations of the female bladder, sagittal section of the pelvis. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

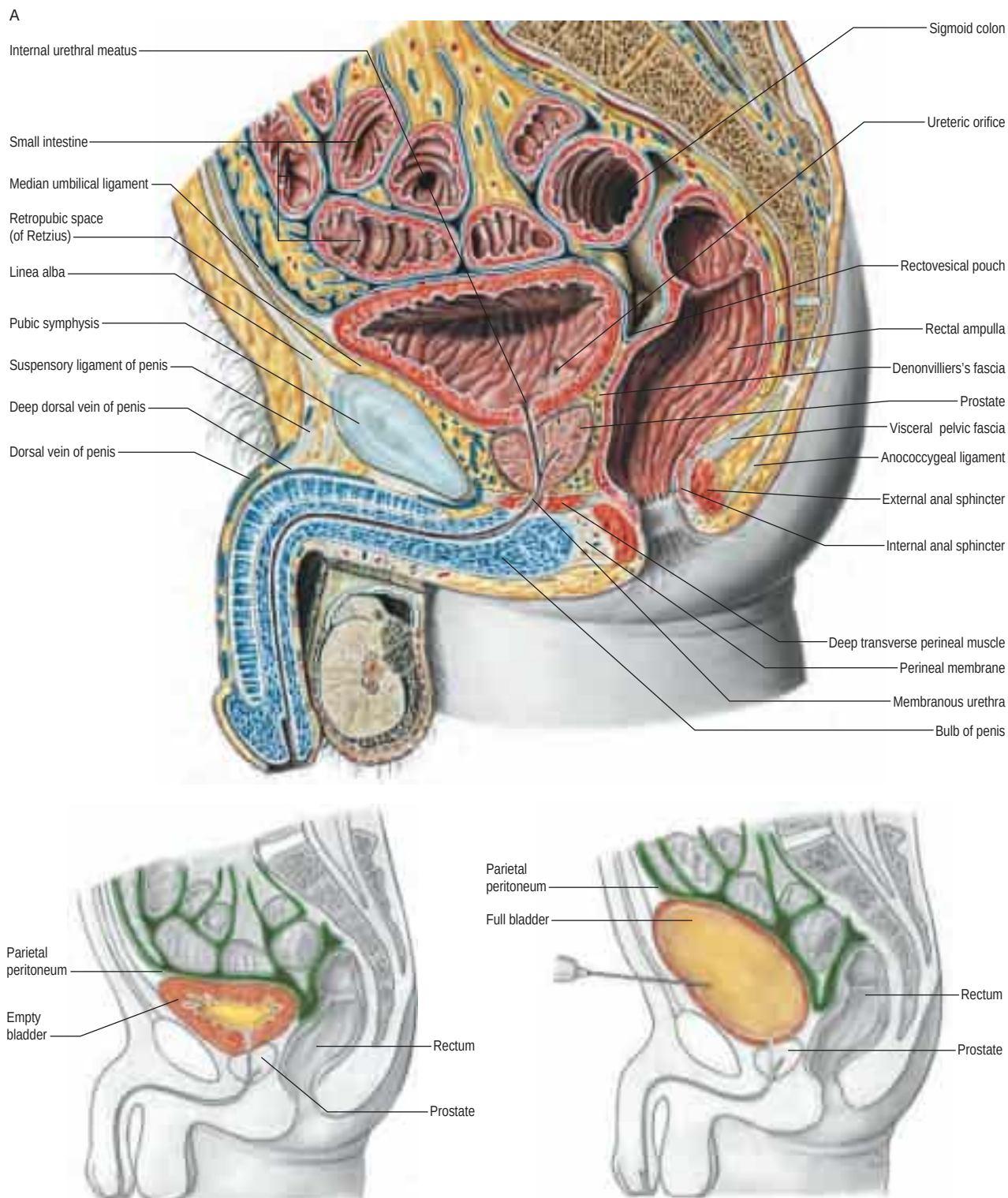


Fig. 75.2 A, The relationship of the bladder and prostate: sagittal section, male pelvis. The relationship of the bladder to the peritoneum and anterior abdominal wall on filling allows suprapubic cystostomy without intraperitoneal urinary leak. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Continued

As the bladder fills, it becomes ovoid (Fig. 75.5; see Fig. 75.2 and Video 75.1). Anteriorly, it displaces the parietal peritoneum from the suprapubic region of the abdominal wall. Its inferolateral surfaces become anterior and rest against the abdominal wall without intervening peritoneum for a distance above the pubic symphysis that varies with the degree of distension, but is commonly 5–7 cm. The distended bladder may be punctured just above the pubic symphysis without traversing the peritoneum (suprapubic cystostomy) (see Fig. 75.2); surgical access to the bladder through the anterior abdominal wall is usually by this route. The summit of the full bladder points up and forwards above the attachment of the median umbilical ligament, so that the peritoneum forms a supravescical recess of varying depth

between the summit and the anterior abdominal wall; this recess often contains coils of small intestine. At birth, the bladder is higher than in the adult because the true pelvis is shallow, and the internal urethral orifice is level with the upper symphyseal border. The bladder is then abdominal rather than pelvic, and extends about two-thirds of the distance towards the umbilicus.

LIGAMENTS OF THE BLADDER

The bladder is anchored inferiorly to the pubis, lateral pelvic side walls and rectum by condensations of pelvic fascia; although these

Urine samples may, therefore, be obtained in children by performing suprapubic needle puncture. A normogram of bladder volume index (BVI=length×width×depth of bladder), based on sonographic measurements in children, is shown in [Figure 75.6](#). The bladder progressively descends with growth, and reaches the adult position shortly after puberty.

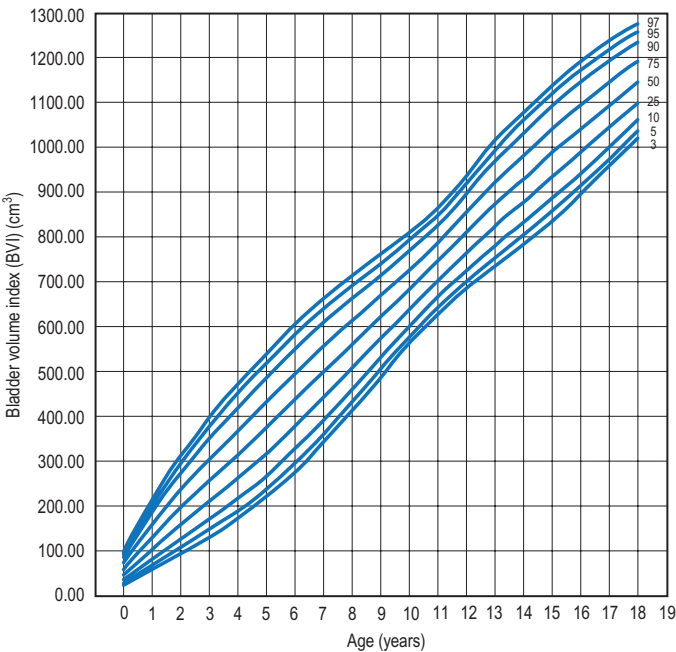


Fig. 75.6 Centiles of bladder volume index in children. (Redrawn with permission from Leung VY, Chu WC, Yeung CK et al Nomograms of total renal volume, urinary bladder volume and bladder wall thickness index in 3,376 children with a normal urinary tract, *Pediatric Radiology* (2007); 37:181–188.)



Fig. 75.2, cont'd B, A sagittal T2-weighted magnetic resonance image of the male pelvis, showing the bladder, prostate and relations.

condensations are not true anatomical ligaments, the term is applied in routine clinical use (see p. 1224 for a further description of the visceral pelvic fascia).

In both sexes, stout bands of fibromuscular tissue, the pubovesical ligaments, extend from the bladder neck to the inferior aspect of the pubic bones; they lie on each side of the median plane, leaving a midline hiatus through which numerous small veins pass. The pubovesical ligaments are derived from the detrusor muscle, part of the detrusor apron (Fig. 75.7). In the female, they constitute the superior extensions of the pubourethral ligaments. In the male, the detrusor apron is described as an extension of detrusor that extends over the anterior surface of the prostate, and condenses distally and anteriorly to form the puboprostatic ligaments.

Other ligaments that have been described in relation to the base of the urinary bladder are the lateral, sacrogenital/uterosacral and cardinal ligaments. The literature is sometimes confusing and even contradictory. The lateral ligament was described by Miles in 1908; although never described in anatomical cadaveric dissection studies, it is recognized clinically as an important structure in the pararectal space at operation. It is a broad band of dense connective tissue, varying in depth from 5 to 7 cm, and passing between the lateral wall of the pelvis and the base of the bladder at the point where the ureter terminates. It contains the middle rectal artery and lymphatic vessels that pass from the lower rectum to the iliac lymph nodes (Takahashi et al 2000). The apex of the bladder is connected to the umbilicus by the remains of the urachus, which forms the median umbilical ligament (see Fig. 75.3; Video 75.2). Composed of longitudinal muscle fibres derived from the detrusor, it becomes more fibrous towards the umbilicus. It usually maintains a lumen lined with epithelium that persists into adult life but is only rarely complicated by a urachal cyst, sinus, fistula or adenocarcinoma.

From the superior surface of the bladder, the peritoneum is carried off in a series of folds: the 'false' ligaments of the bladder. Anteriorly, there are three folds (Fig. 75.8): the median umbilical fold over the median umbilical ligament (urachus), and two medial umbilical folds over the obliterated umbilical arteries. The inferior epigastric vessels (Fig. 75.9) are lateral to these folds on the anterior abdominal wall and are termed lateral umbilical ligaments in descriptions of anterior abdominal wall anatomy.

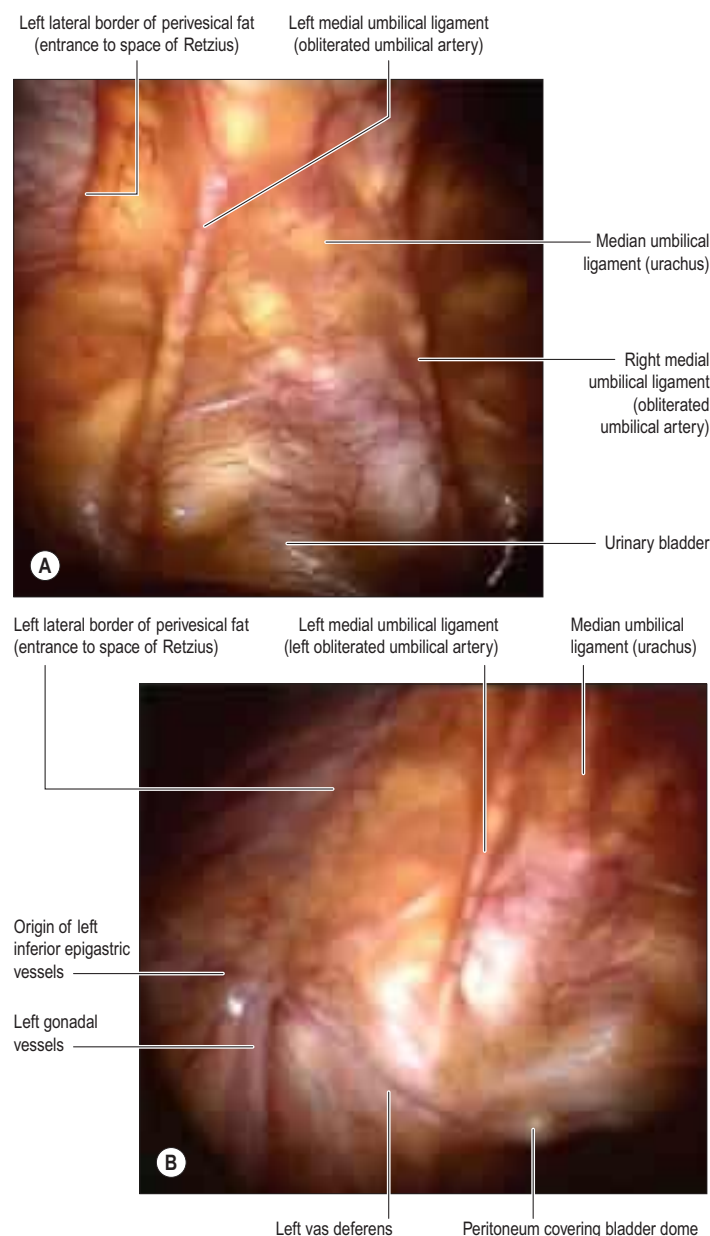


Fig. 75.3 **A**, A view of the midline anterior abdominal wall and ligaments during laparoscopy. **B**, A view of the left side of the anterior abdominal wall and ligaments during laparoscopy.

BLADDER INTERIOR

Vesical mucosa

Almost all of the vesical mucosa (Fig. 75.10) is attached only loosely to subjacent muscle; it folds when the bladder empties, and the folds are stretched flat as it fills. Over the trigone, immediately above and behind the internal urethral orifice, it is adherent to the subjacent muscle layer and is always smooth. The anteroinferior angle of the trigone is formed by the internal urethral orifice, its posterolateral angles by the ureteric orifices. The superior trigonal boundary is a slightly curved inter-ureteric bar, which connects the two ureteric orifices and is produced by the continuation into the vesical wall of the ureteric internal longitudinal muscle. Laterally, this ridge extends beyond the ureteric openings as ureteric folds, produced by the terminal parts of the ureters, which run obliquely through the bladder wall. At cystoscopy, the inter-ureteric crest appears as a pale band and is a guide to the ureteric orifices (Fig. 75.10B).

Trigone

The smooth muscle of the trigone consists of two distinct layers, sometimes termed the superficial trigonal muscle and deep trigonal detrusor muscle. The latter is composed of muscle cells, indistinguishable from those of the detrusor, and is simply the posteroinferior portion of the detrusor muscle proper. The superficial trigonal muscle represents a morphologically distinct component of the trigone, which, unlike the detrusor, is composed of relatively small-diameter muscle bundles that

The urachus may play a critical role in maintaining fetal life when atresia of the urethra results in complete obstruction to the flow of amniotic fluid. Anhydramnios noted at 17 weeks' gestation has been reported to resolve by 21 weeks when the patent urachus acts as a fistula between the bladder and the amniotic space, preserving pulmonary and renal function (Stalberg and Gonzalez 2012).

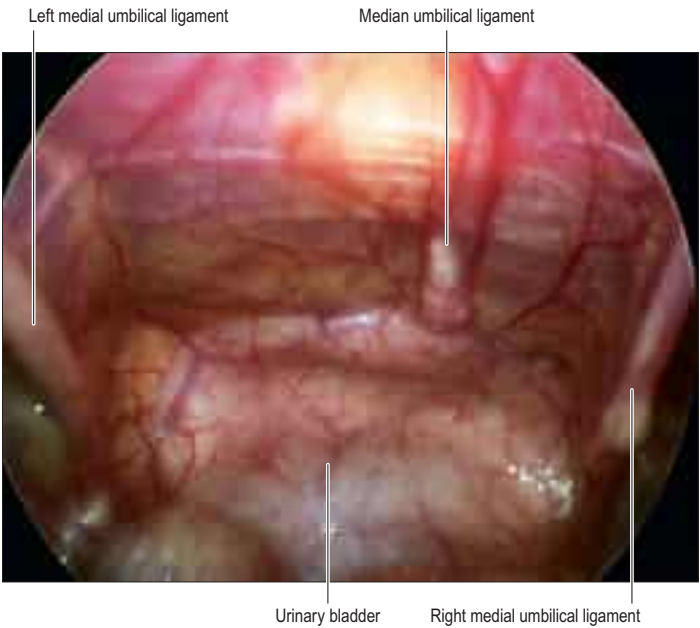


Fig. 75.8 A laparoscopic view of the empty bladder and median and medial umbilical folds in a 12-month-old boy. (Courtesy of Mr Girish Jawaheer.)

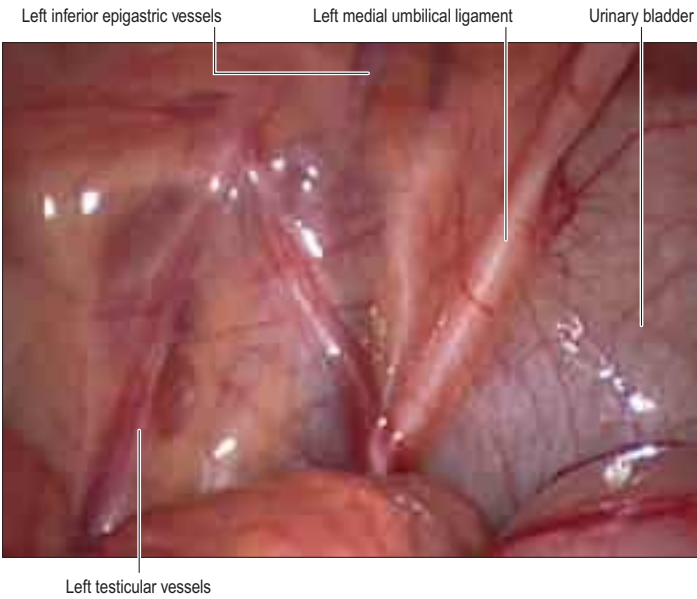


Fig. 75.9 A laparoscopic view of the left inferior epigastric vein (lateral umbilical ligament) in a 12-month-old boy. (Courtesy of Mr Girish Jawaheer.)

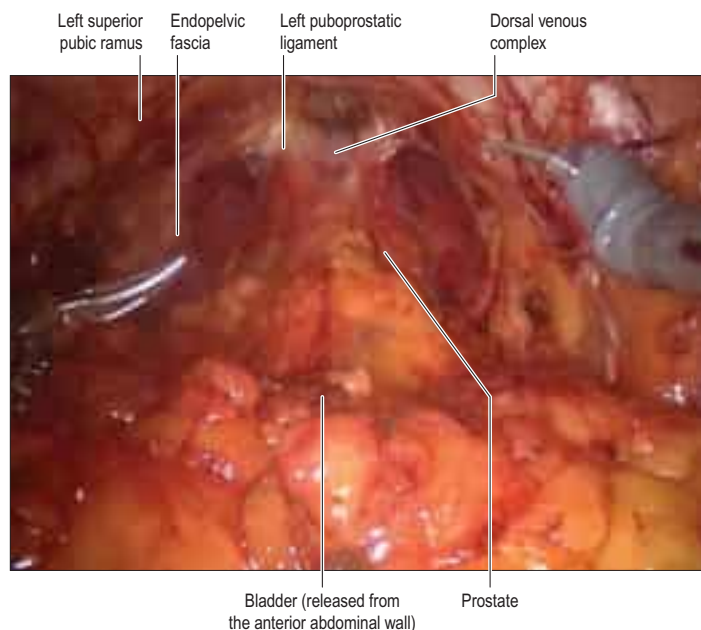


Fig. 75.4 A transperitoneal view of the male pelvis, as seen during robotic-assisted laparoscopic radical prostatectomy. The bladder is released from the anterior abdominal wall, the space of Retzius is developed and the endopelvic fascia opened.

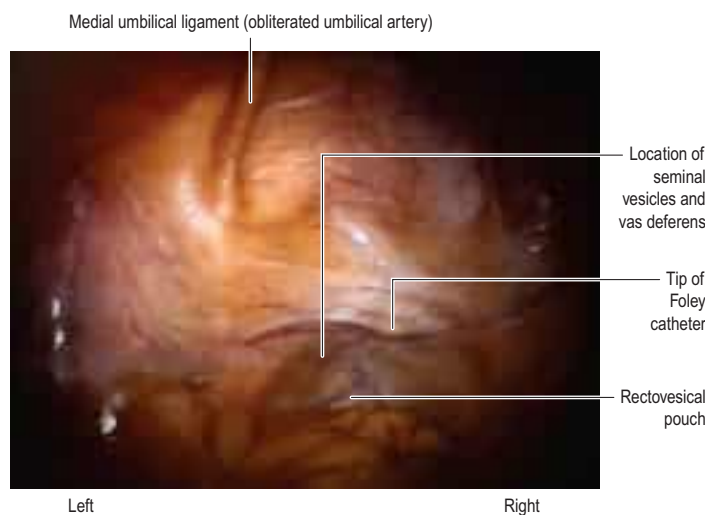


Fig. 75.5 Laparoscopic view of the full bladder.

are continuous proximally with those of the intramural ureters. The superficial trigonal muscle is relatively thin but is generally described as becoming thickened along its superior border to form the interureteric ridge (bar). Similar thickenings occur along the lateral edges of the superficial trigone. In both sexes, the superficial trigone muscle becomes continuous with the smooth muscle of the proximal urethra, and extends in the male along the urethral crest as far as the openings of the ejaculatory ducts.

Ureteric orifices

The slit-like ureteric orifices are placed at the posterolateral trigonal angles (Fig. 75.10A,B). In empty bladders, they are approximately 2.5 cm apart, and 2.5 cm from the internal urethral orifice; in distension, these measurements may be doubled.

Internal urethral orifice

The internal urethral orifice is sited at the trigonal apex, the lowest part of the bladder, and is usually somewhat crescentic in section. There is often an elevation immediately behind it in adult males (particularly past middle age), which is caused by the median prostatic lobe.

BLADDER NECK

The smooth muscle of the bladder neck is histologically, histochemically and pharmacologically distinct from the detrusor muscle proper and so the bladder neck should be considered as a separate functional

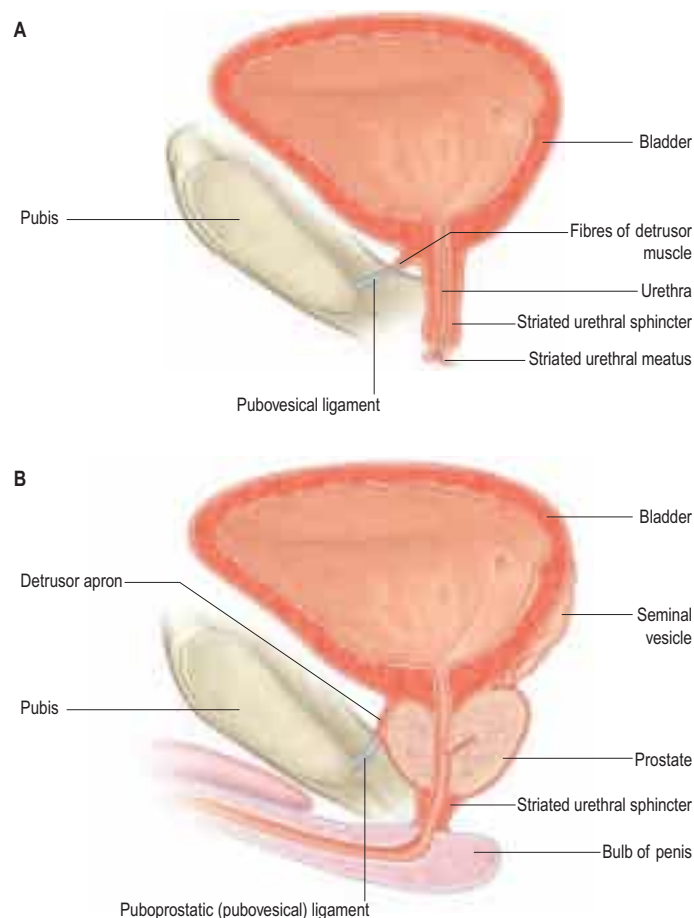


Fig. 75.7 Ligaments that anchor the neck of the bladder and pelvic part of the urethra to the pelvic bone. **A**, In the female. **B**, In the male. In males, the detrusor extends over the anterosuperior aspect of the prostate and inserts into the pubic bone. It condenses to form pubovesical/puboprostatic ligaments. (A–B, With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

unit. The arrangement of smooth muscle in this region is quite different in males and females, and therefore will be described separately.

Female

In the female, the bladder neck consists of morphologically distinct smooth muscle (Fig. 75.10C). The large-diameter fasciculi characteristic of the detrusor are replaced in the region of the bladder neck by small-diameter fasciculi that extend obliquely or longitudinally into the urethral wall. In the normal female, the bladder neck sits above the pelvic floor, supported predominantly by the pubovesical ligaments (see Fig. 75.7A), the endopelvic fascia of the pelvic floor and levator ani. These support the urethra at rest; with elevated intra-abdominal pressure, the levators contract, increasing urethral closure pressure to maintain continence. This anatomical arrangement commonly alters after parturition and with increasing age, such that the bladder neck lies beneath the pelvic floor, particularly when intra-abdominal pressure rises, which means that the mechanism described above fails to maintain continence and women may experience stress incontinence (Klutke and Siegel 1995).

Male

In the male, the bladder neck is completely surrounded by a circular collar of smooth muscle, with its own distinct adrenergic innervation, which extends distally to surround the preprostatic portion of the urethra. These smooth muscle bundles are distinct from the smooth muscle bundles that run in continuity from the bladder neck down to the prostatic urethra, and from the smooth muscle within the prostate. The bundles that form this 'preprostatic sphincter' are small in size compared with the muscle bundles of the detrusor, and are separated by a relatively larger connective tissue component rich in elastic fibres.

The bladder neck is sometimes called the proximal or internal sphincter mechanism, to distinguish it from the distal urinary sphincter mechanism. The internal sphincter contributes to urinary continence and, in the face of distal sphincteric incompetence, can, at times, maintain continence independently. Unlike the detrusor and the rest of the

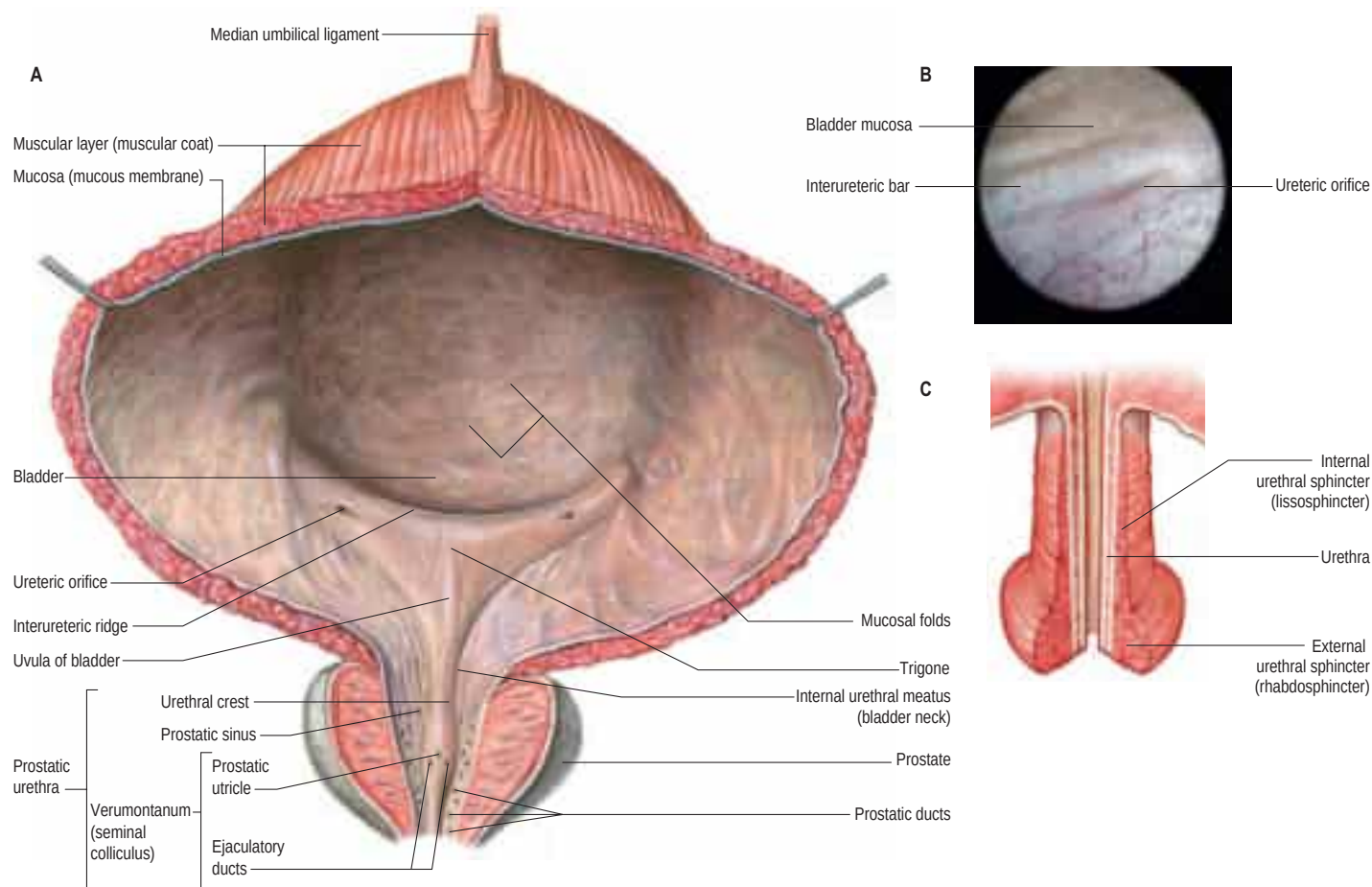


Fig. 75.10 **A**, A coronal section of the urinary bladder in the male. The mucosal folds are dependent on the state of filling. **B**, The ureteric orifice seen at endoscopy. **C**, A posterior view of the bladder neck in the female. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

urethral smooth muscle (common to both sexes), the preprostatic sphincter is richly supplied with sympathetic noradrenergic nerves and is almost totally devoid of parasympathetic cholinergic nerves (see below).

Bladder outflow obstruction

In progressive chronic obstruction to micturition, e.g. as a result of prostatic enlargement or urethral stricture, or in children with congenital bladder outflow obstruction, e.g. posterior urethral valves, the muscle of the bladder hypertrophies. The muscle fasciculi increase in size and, because they interlace in all directions, a thick-walled 'trabeculated bladder' is produced. Mucosa between the fascicles forms 'diverticula'. When outflow is obstructed, emptying is not complete; some urine remains and may become infected. Back-pressure from a chronically distended bladder may gradually dilate the ureters, renal pelves and even the renal collecting tubules, which can result in progressive renal impairment.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The bladder is supplied principally by the superior and inferior vesical arteries (see Figs 77.3, 74.22), derived from the anterior trunk of the internal iliac artery, and supplemented by the obturator and inferior gluteal arteries. In the female, additional branches are derived from the uterine and vaginal arteries.

Superior vesical artery

The superior vesical artery supplies many branches to the fundus of the bladder. The artery to the vas deferens often originates from one of these and accompanies the vas deferens to the testis, where it anastomoses with the testicular artery. Other branches supply the ureter. The initial portion of the superior vesical artery is the proximal, patent, section of the fetal umbilical artery.

Inferior vesical artery

The inferior vesical artery often arises with the middle rectal artery from the internal iliac artery. It supplies the base of the bladder, prostate, seminal vesicles and lower ureter, and may sometimes provide the artery to the vas deferens. Prostatic branches communicate across the midline.

Veins

The veins that drain the bladder form a complicated plexus on its inferolateral surfaces and pass backwards in the lateral ligaments of the bladder to end in the internal iliac veins (see Fig. 77.3).

Lymphatic drainage

Lymphatics that drain the bladder begin in mucosal, intermuscular and serosal plexuses (see Fig. 77.3). There are three sets of collecting vessels; most end in the external iliac nodes. Vessels from the trigone emerge on the exterior of the bladder to run superolaterally. Vessels from the superior surface of the bladder converge to the posterolateral angle and pass superolaterally to the external iliac nodes (some may go to the internal or common iliac group). Vessels from the inferolateral surface of the bladder ascend to join those from the superior surface or run to the lymph nodes in the obturator fossa. Minute nodules of lymphoid tissue may occur along the vesical lymph vessels.

INNERVATION

The nerves supplying the bladder arise from the pelvic plexuses, a mesh of autonomic nerves and ganglia lying on the lateral aspects of the rectum, internal genitalia and bladder base (Fig. 75.11). They consist of both sympathetic and parasympathetic components, each of which contains both efferent and afferent fibres. For further reading, see Mundy et al (1999a).

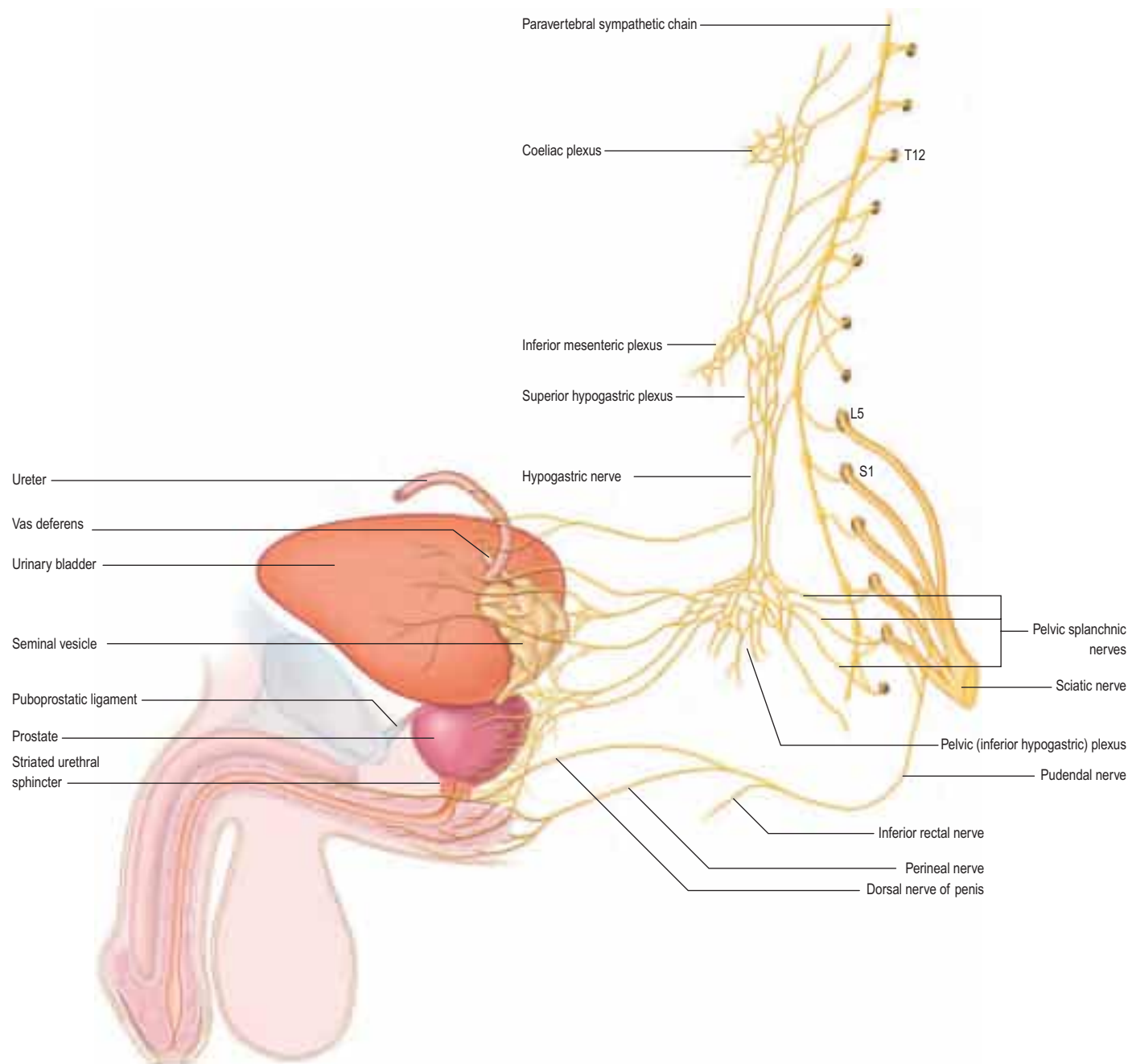


Fig. 75.11 Innervation of the lower urinary tract and male genitalia. (Redrawn with permission from Dyck P, Thomas PK, 2005, *Peripheral Neuropathy*, Saunders, Elsevier.)

Efferent fibres

Parasympathetic fibres arise from the second to the fourth sacral segments of the spinal cord and enter the pelvic plexuses on the posterolateral aspects of the rectum as the pelvic splanchnic nerves. The sympathetic fibres are derived from neuronal cell bodies in the lower three thoracic and upper two lumbar segments of the spinal cord and form the coeliac and mesenteric plexuses around the great vessels in the abdomen; from here, the hypogastric plexuses descend into the pelvis as fairly discrete nerve bundles within the extraperitoneal connective tissue posterior to the ureter on each side. The anterior part of the pelvic plexus is known as the vesical plexus. Small groups of autonomic neurones occur within the plexus and throughout all regions of the bladder wall. These multipolar intramural neurones are rich in acetylcholinesterase (AChE) and occur in ganglia consisting of up to 20 neuronal cell bodies. The majority of the preganglionic nerve terminals correspond morphologically to presumptive cholinergic fibres. Noradrenergic terminals also relay on cell bodies in the pelvic plexus; it is not known whether similar nerves synapse on intramural bladder ganglia.

The urinary bladder (including the trigonal detrusor muscle) is profusely supplied with nerves, which form a dense plexus among the detrusor muscle cells. The majority of these nerves contain AChE and occur in abundance throughout the muscle coat of the bladder. Axonal varicosities adjacent to detrusor muscle cells possess features that are

considered to typify cholinergic nerve terminals, and contain clusters of small (50 nm diameter) agranular vesicles, occasional large (80–160 nm diameter) granulated vesicles and small mitochondria. Terminal regions approach to within 20 nm of the surface of the muscle cells and may be partially surrounded by Schwann cell cytoplasm; more often, they are naked nerve endings. The human detrusor muscle possesses a sparse but definite supply of sympathetic noradrenergic nerves that generally accompany the vascular supply and only rarely extend among the myocytes. Non-adrenergic, non-cholinergic nerves have been identified, and a number of other neurotransmitters or neuromodulators have been detected in intramural ganglia, including the peptide somatostatin. The superficial trigonal muscle is associated with more noradrenergic (sympathetic) fibres than cholinergic (parasympathetic) nerves, a difference that supports the view that the superficial trigonal muscle should be regarded as 'ureteric' rather than 'vesical' in origin. It is, however, important to emphasize that the superficial trigonal muscle forms a very minor part of the total muscle mass of the bladder neck and proximal urethra in either sex and is probably of little significance in the physiological mechanisms that control these regions.

The smooth muscle of the bladder neck in males is predominantly orientated obliquely or circularly. It is sparsely supplied with cholinergic (parasympathetic) nerves but possesses a rich noradrenergic (sympathetic) innervation. A similar distribution of autonomic nerves also

occurs in the smooth muscle of the prostate gland, seminal vesicles and vasa deferentia. Stimulation of sympathetic nerves causes contraction of smooth muscle in the wall of the genital tract, resulting in seminal emission. Concomitant sympathetic stimulation of the proximal urethral smooth muscle causes sphincteric closure of the preprostatic sphincter, thereby preventing reflux of ejaculate into the bladder.

It is extensively disrupted in the vast majority of men undergoing bladder neck surgery, e.g. transurethral resection of the prostate, which may result in retrograde ejaculation. Similarly, anejaculation can result when the sympathetic fibres are disrupted during retroperitoneal lymph node dissection surgery.

Although this genital function of the bladder neck of the male is well established, it is not known whether the smooth muscle of this region plays an active role in maintaining urinary continence. In contrast, the smooth muscle of the bladder neck of the female receives relatively few noradrenergic nerves but is richly supplied with presumptive cholinergic fibres. The sparse supply of sympathetic nerves presumably relates to the absence of a functioning 'genital' portion of the wall of the female urethra.

The lamina propria of the fundus and inferolateral walls of the bladder is virtually devoid of autonomic nerve fibres, apart from some noradrenergic and occasional presumptive cholinergic perivascular nerves. However, the density of nerves unrelated to blood vessels increases closer to the urethral orifice. At the bladder neck and trigone, a nerve plexus of cholinesterase-positive axons extends throughout the lamina propria, independent of blood vessels. Some of the larger-diameter axons are myelinated and others lie adjacent to the basal urothelial cells. In the absence of any obvious effector target sites, the subepithelial nerve plexuses of the bladder and the ureter are assumed to subserve a sensory function.

Afferent fibres

Vesical nerves are also concerned with pain and awareness of distension, and are stimulated by distension or spasm due to a stone, inflammation or malignant disease; they travel in sympathetic and parasympathetic nerves, but predominantly in the latter. Division of the sympathetic paths (e.g. 'presacral neurectomy') or of the superior hypogastric plexus, therefore, does not materially relieve vesical pain, whereas considerable relief follows bilateral anterolateral cordotomy. Since nerve fibres mediating awareness of distension travel in the posterior columns (fasciculus gracilis), the patient still retains awareness of the need to micturate after anterolateral cordotomy. The nerve endings detecting noxious stimuli are probably of more than one type; a subepithelial plexus of fibres containing dense vesicles, which are probably afferent endings, has been described.

MICROSTRUCTURE

The bladder consists of four layers: a lining epithelium (urothelium), lamina propria, muscularis propria and serosa.

Lining epithelium or urothelium

Urothelium (transitional epithelium; see Fig. 2.5D) is 4–7 cells thick; it may appear to be attenuated to 2–3 cells thick when the bladder is fully distended. It contains three distinctive cell layers: a basal layer, an intermediate layer and a superficial ('umbrella' cell) layer (see Fig. 75.15). The basal layer consists of small cuboidal cells from which the upper layers arise. The intermediate layers are polygonal and possess the capacity to stretch and flatten. The superficial layer forms a protective, almost impermeable, surface for the bladder mucosa and consists of large, sometimes multinucleated, cells displaying degenerative changes in their cytoplasm; these cells are ultimately exfoliated into the urine. The apical surface of the umbrella cell layer is covered by 16 nm protein particles packed hexagonally to form two-dimensional crystals of asymmetric unit membranes (AUMs), which contribute to the permeability barrier function of the urinary bladder, preventing reabsorption of urine across the urothelium into the blood stream. Islands or nests of urothelium (von Brunn's nests) may become separated from the surface during development and are found embedded in the underlying lamina propria; they may undergo central degeneration to form cysts (cystitis cystica).

Lamina propria

The lamina propria forms a connective tissue bed supporting the overlying urothelium, from which it is separated by a basement membrane. It is rich in capillaries, lymphatics and nerve endings, and contains elastic fibres and a thin, poorly defined, layer of smooth muscle fibres, the muscularis mucosae, which is variably distributed within the

bladder. This layer is often very superficial, lying close beneath the urothelium, but is sometimes deeper in the lamina propria, where it may be well developed; it must be distinguished from the muscularis propria. Several functions have been proposed for the lamina propria, including acting as the capacitance layer of the bladder (determining bladder compliance and enabling adaptive changes to increasing volumes); providing factors that influence the growth of the urothelium and/or detrusor; and playing a role in signal transduction (Andersson and McCloskey 2014).

Interstitial cells of Cajal-like cells

Interstitial cells of Cajal (ICC) are a specialized population of cells involved in smooth muscle excitability that were initially described in the gastrointestinal tract (p. 1043). Cells with a similar morphology but expressing platelet-derived growth receptor alpha (PDGFR α), rather than the tyrosine kinase receptor Kit that typifies ICC, and therefore designated interstitial cells of Cajal-like cells (ICC-LC), have been reported in the urethra, vas deferens, prostate, bladder, corpus cavernosum, ureter, Fallopian tube, oviduct and uterus, where they form a functional syncytium with nerves and smooth muscle cells. They may function as electrical pacemakers in the urethra and prostate. In the bladder, they are found closely apposed to detrusor smooth muscle cells, in the lamina propria and within the suburothelium; it has been tentatively suggested that one of their functions in the bladder is to act as intermediary cells transducing nerve signals to detrusor smooth muscle cells (Drumm et al 2014).

Muscularis propria

The muscularis propria is the detrusor muscle of the bladder wall. It consists of three indistinct layers: an inner and outer longitudinal, and an intermediate circular layer. In contrast to the muscularis mucosae, which consists of small wispy fibres, the muscularis propria is made up of larger, poorly defined, bundles of smooth muscle that form well-defined inner circular and outer longitudinal layers at the neck of the bladder.

Serosa

The serosa partially covers the bladder. It is lined externally by mesothelium, beneath which there is a variable amount of vascularized adipose tissue that frequently extends into the muscularis propria and, occasionally, into the lamina propria.

MALE URETHRA

The male urethra (Fig. 75.12, see Fig. 75.2) is 18–20 cm long, and extends from the internal orifice in the urinary bladder to the external opening, or meatus, at the end of the penis. It may be considered in two parts. The anterior urethra is approximately 16 cm long and lies within the perineum (proximally) and the penis (distally), surrounded by the corpus spongiosum. The posterior urethra is 4 cm long and lies in the pelvis proximal to the corpus spongiosum, where it is acted on by the urogenital sphincter mechanisms. Functionally, both parts act as a conduit.

The anterior urethra is subdivided into a proximal component, the bulbar urethra, which is surrounded by bulbospongiosus and entirely within the perineum, and a pendulous or penile component, which continues to the tip of the penis. The posterior urethra is divided into preprostatic, prostatic and membranous segments. In the flaccid penis, the urethra has a double curve. The urethral canal is a mere slit, except during the passage of fluid; in transverse section, it is transversely arched in the prostatic part, stellate in the preprostatic and membranous portions, transverse in the bulbar and penile portions, and sagittal at the external orifice.

Posterior part Preprostatic urethra

The preprostatic urethra is approximately 1 cm in length, and extends from the base of the bladder to the prostate (Fig. 75.12). Small periurethral glands at this site may contribute to benign prostatic hyperplasia (BPH) and symptoms of outflow obstruction in older men.

Prostatic urethra

The prostatic urethra is 3–4 cm in length and tunnels through the substance of the prostate, closer to the anterior than the posterior surface of the gland (Fig. 75.13; see Fig. 75.10). It is continuous above with the preprostatic part and emerges from the prostate slightly anterior to its apex (the most inferior point of the prostate). Throughout most of its length, the posterior wall possesses a midline ridge, the urethral crest, that projects into the lumen, causing it to appear



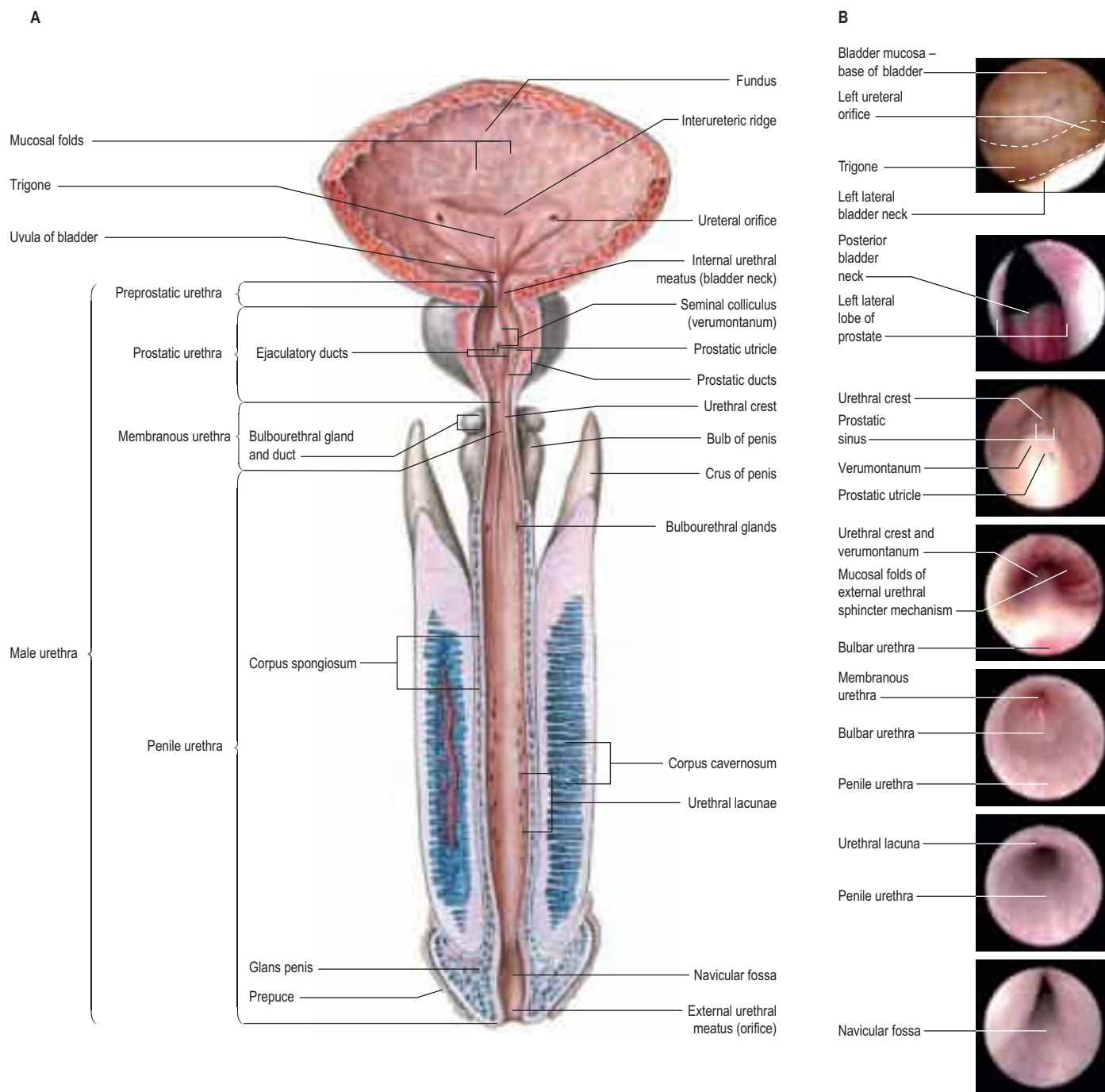


Fig. 75.12 A, The male urethra. **B**, Endoscopic views (top to bottom): posterior bladder wall and trigone, bladder neck, urethral crest, membranous urethra, bulbar urethra, penile urethra, navicular fossa. (A, with permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

crenate in transverse section. On each side of the crest, the floor of a shallow depression, the prostatic sinus, is perforated by the orifices of 15–20 prostatic ducts. An elevation, the verumontanum (seminal colliculus), occurs at about the middle of the length of the urethral crest; at this point, the urethra turns anteriorly and contains the slit-like orifice of the prostatic utricle. The extent of angulation most often approximates 30–35° but can change with age and differs between individuals. The verumontanum is used as a surgical landmark for the urethral sphincter during transurethral resection for benign enlargement of the prostate.

The two small openings of the ejaculatory ducts open on both sides of, or just within, the prostatic utricle, a cul-de-sac approximately 6 mm long, that runs upwards and backwards in the substance of the prostate behind its median lobe. Its walls are composed of fibrous tissue, muscular fibres and mucous membrane, the latter pitted by the openings of numerous small glands. It develops from the paramesonephric ducts or urogenital sinus, and has been thought to be homologous with the vagina of the female so that it is sometimes called the 'vagina masculina'; the more usual view is that it is a uterine homologue – hence the name 'utricle'.

The lowermost part of the prostatic urethra is fixed by the puboprostatic ligaments and is therefore immobile.

Membranous urethra

The membranous part of the urethra is the shortest (2–2.5 cm), least dilatable and, with the exception of the external orifice, the narrowest section of the urethra. It descends with a slight ventral concavity from the prostate to the bulb of the penis, passing through the perineal membrane, 2.5 cm postero-inferior to the pubic symphysis. The wall of the membranous urethra consists of a muscle coat that is separated from the epithelial lining by a narrow layer of fibroelastic connective tissue. The muscle coat contains a relatively thin layer of bundles of smooth muscle, which are continuous proximally with those of the prostatic urethra, and a prominent outer layer of circularly orientated striated muscle fibres, which together form the external urethral sphincter.

Urinary continence at the level of the membranous urethra is mediated by the radial folds of urethral mucosa, the submucosal connective tissue, the intrinsic urethral smooth muscle, the striated muscle fibres and the pubourethral component of levator ani. The muscle coat of the urethra and puborectalis surround the membranous urethra and are attached to the inner surface of the ischiopubic ramus; fibres also reach up to the lowest part of the neck of the bladder and lie on the surface of the prostate. The striated external urethral sphincter has a posterior fibrous defect and is inserted throughout its length into the perineal

Cystic dilation of the prostatic utricle ([Fig. 75.14](#)) may present in childhood with urinary tract infections, recurrent epididymitis or haematuria.



Fig. 75.14 Cystic dilation of the prostatic utricle (arrow) demonstrated in an 8-month-old child during a micturating cystourethrogram. (Courtesy of Mr Alok Godse.)

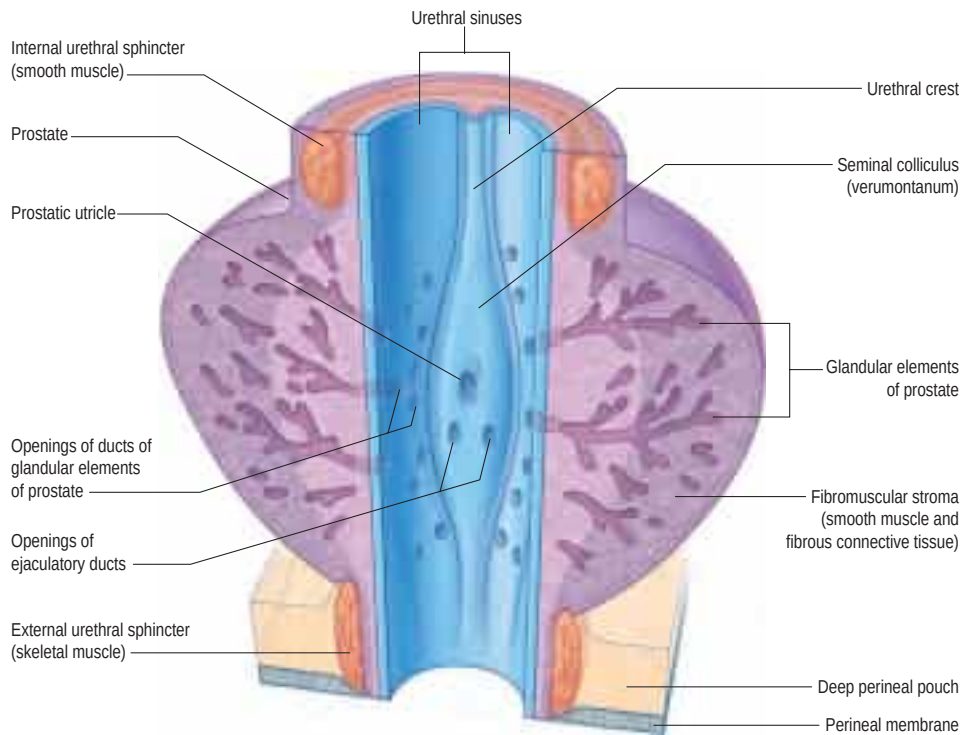


Fig. 75.13 The prostatic part of the male urethra. The raised part of the urethral crest is the seminal colliculus, known clinically as the verumontanum. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

body. It is related to the dorsal vein complex anteriorly, levator ani laterally, and the perineal body and rectourethralis posteriorly; it is suspended from the pubis by fibrous tissue that extends from its anterior and lateral parts to the puboprostatic ligaments posteriorly and to the suspensory ligament of the penis anteriorly. The bulbourethral glands are invested in sphincteric muscle and drain into the membranous urethra during sexual excitement.

Anterior part

The anterior, or spongiosum, part of the urethra lies within the corpus spongiosum penis (see Fig. 76.3). In the flaccid penis, it is about 15 cm long and extends from the end of the membranous urethra to the external urethral orifice on the glans penis. It starts below the perineal membrane at a point anterior to the lowest level of the pubic symphysis as the bulbar urethra, the widest part of the urethra, surrounded by bulbospongiosus. The bulbourethral glands open into the bulbar urethra approximately 2.5 cm below the perineal membrane. The urethra next curves downwards as the penile urethra. It is a narrow, transverse slit when empty, and has a diameter of approximately 6 mm when passing urine. It is dilated at its termination within the glans penis, where it is known as the navicular fossa. The external urethral orifice is the narrowest part of the urethra and, in the adult, is a sagittal slit, about 6 mm long, bounded on each side by a small labium.

The urethral epithelium, particularly in the bulbar and distal penile segments, presents the orifices of numerous small mucous urethral glands that lie in the submucosa. It also contains a number of small pit-like recesses, or lacunae, of varying sizes with orifices directed forwards. One, the lacuna magna, is larger than the rest and is situated in the roof of the navicular fossa.

Traumatic injury to the male urethra



Available with the *Gray's Anatomy* e-book

Congenital anomalies of the male urethra



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VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Urethral artery

The urethral artery arises from the internal pudendal artery or common penile artery just below the perineal membrane, and travels through the corpus spongiosum to reach the glans penis. It supplies the urethra and the erectile tissue around it. The urethra is also supplied by the dorsal penile artery, via its circumflex branches on each side and in a retrograde fashion from the glans, by its terminal branches. The blood supply

through the corpus spongiosum is so plentiful that the urethra can be divided without compromising its vascular supply.

Veins

The venous drainage of the anterior urethra is to the dorsal veins of the penis and internal pudendal veins, which drain to the prostatic plexus. The posterior urethra drains into the prostatic and vesical venous plexuses, which drain into the internal iliac veins.

Lymphatic drainage

Vessels from the posterior urethra pass mainly to the internal iliac nodes; a few may end in the external iliac nodes. Vessels from the membranous urethra accompany the internal pudendal artery. Vessels from the anterior urethra accompany those of the glans penis, most ending in the deep inguinal nodes; some may end in superficial nodes, while others may traverse the inguinal canal to end in the external iliac nodes.

INNERVATION

The prostatic plexus supplies the smooth muscle of the prostate and prostatic urethra (see Fig. 75.11). On each side, it is derived from the pelvic plexus and lies on the posterolateral aspect of the seminal vesicle and prostate. The cavernous (deep, cavernosal) nerves, both major and minor, pierce the bulb of the corpus spongiosum to innervate the cavernous bodies of the penis. The sympathetic autonomic nerves that supply the internal urethral sphincter are derived from the pelvic plexus as it descends in the pelvis adjacent to the inferior prostatovesical pedicle and prostate, and function to prevent retrograde ejaculation. Parasympathetic preganglionic axons arise from neuronal cell bodies in the second to fourth sacral spinal segments. The nerve supply of the external urethral sphincter is controversial. It is generally believed to be supplied by neurones in Onuf's nucleus and by perineal branches of the pudendal nerve lying on the perineal aspect of the pelvic floor; in both instances, the axons arise from neurones in the second to fourth sacral spinal segments. Fibres from Onuf's nucleus (somatic) travel with the pelvic plexus on each side until they branch off to run on the pelvic aspect of the pelvic floor to enter the membranous urethra.

MICROSTRUCTURE

The epithelium lining the preprostatic urethra and the proximal part of the prostatic urethra is a typical urothelium (Fig. 75.15). It is continuous with that lining the bladder, and with the epithelium lining the ducts of the prostate and bulbourethral glands, the seminal vesicles, and the vasa deferentia and ejaculatory ducts. These relationships are important in the spread of urinary tract infections.

The mean length of the slit is 5.4 mm in boys between the ages of 0.3 and 15 years (Hutton and Babu 2007). The width of the external urethral orifice in uncircumcised boys between the ages of 5 months and 16 years varies from 3.5 to 7.5 mm (Orkiszewski and Madej 2010).

The urethra may be ruptured by a fall-astride (straddle) injury to the bulbar urethra in the perineum, or by an injury related to a pelvic fracture. These injuries usually affect the junction of the membranous with the bulbar segments across the perineal membrane. One of the complications associated with such injuries is extravasation of urine. After an injury to the bulbar urethra, urine usually extravasates between the perineal membrane and the membranous layer of the superficial fascia (clinically, known as Colles' fascia). As both of these are attached firmly to the ischiopubic rami, extravasated fluid cannot pass posteriorly because the two layers are continuous around the superficial transverse perineal muscles. Laterally, the spread of urine is blocked by the pubic and ischial rami. Urine cannot enter the lesser pelvis through the perineal membrane if this remains intact, and so it tracks anteriorly into the loose connective tissue of the scrotum and penis, and, from there, to the anterior abdominal wall. If the posterior urethra is injured, urine is extravasated into the pelvic extraperitoneal tissue; if the perineal membrane is also torn, then urine may be extravasated into the perineum.

Hypospadias, found in 1 in 300 boys, most often results in the urethra opening in the distal penis, either on the ventral aspect of the penis or, more proximally, on to the perineum. There is also an associated abnormality of the prepuce, which is longer dorsally and lacking ventrally, and often an associated chordee, which causes a ventral curvature of the penis. The microvessel density of the prepuce is reduced in children with hypospadias and this has surgical implications when preputial flaps are used for the repair of hypospadias (Yucel et al 2004, Cagri Savas et al 2011). Anomalies of arterial vascular anatomy of the prepuce have been described in boys with hypospadias; it is, therefore, important for hypospadias to be identified prior to circumcision because the abnormal foreskin is sometimes used for surgical correction of the deformity. The anogenital distance (distance from the anus to the base of penis) is reduced in Caucasian boys with hypospadias (Hsieh et al 2012).

Epispadias occurs in approximately 1 in 100,000 boys; it is typically part of the exstrophy–epispadias complex but may occur in isolation. The urethra is either completely open dorsally or uncovered to the level of the pubic symphysis. The neurovascular bundles are anterolateral along the proximal portions, and lateral along the middle and distal portions, of the corporeal bodies (Hurwitz et al 1986). Other features of epispadias are shortened penis and dorsal chordee; the incidence of cryptorchidism is increased ten-fold.

Posterior urethral valves occur in 1 in 5000–8000 males and are the most common cause of urinary outflow obstruction in male infants. The most common type (type I) is believed to occur if the Wolffian ducts open too anteriorly on to the primitive prostatic urethra; this abnormal migration of the ducts leaves behind thick vestigial tissue that forms rigid valve cusps extending caudally from the verumontanum. Megalourethra may be associated with posterior urethral valves. Congenital anterior urethral valves are a rare cause of urethral obstruction in boys and may be associated with posterior urethral valves and hypospadias.

Very rarely, urethral duplication occurs; the two urethrae almost invariably lie on top of each other rather than side by side. One of the urethrae, usually the more dorsal, may be blind-ending. Congenital prepubic sinus consists of a midline tract in the skin passing from the suprapubic region towards the anterior bladder without communication, and is believed to be a variant of dorsal urethral duplication. Congenital urethrocutaneous fistulae are very rare anomalies in which the urethra opens on the ventral surface of the penis in the absence of chordee or hypospadias.

Congenital rectourethral fistulae may be present in children born with anorectal malformation. The prostatic or bulbar urethra is usually affected, and the rectum and urethra share a common wall immediately above the fistula site.

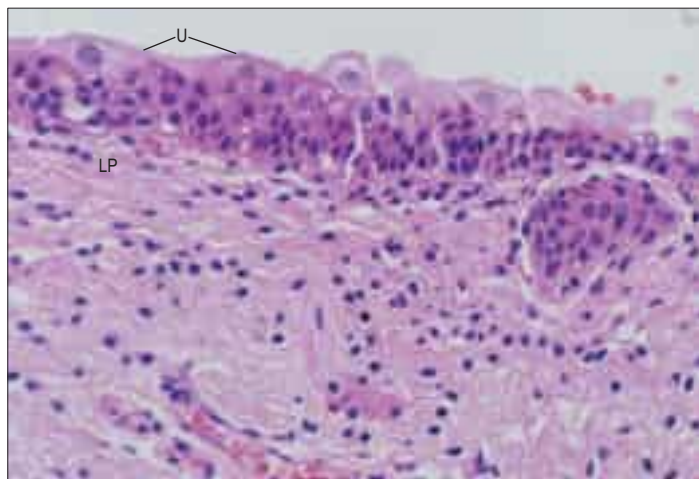


Fig. 75.15 The bladder is lined by urothelium with a surface 'umbrella cell' layer (U) and a layer of intermediate cells (3–5 cells thick). The lamina propria (LP) consists of stroma with blood vessels and von Brunn's nests.

The epithelium changes below the openings of the ejaculatory ducts to a pseudostratified or stratified columnar type, which lines the membranous urethra and the major part of the penile urethra. Mucus-secreting cells are common throughout this epithelium and frequently occur in small clusters in the penile urethra. Branching tubular para-urethral glands secrete protective mucus on to the urethral epithelial lining and are especially numerous on its dorsal aspect. In older men, many of the deep recesses of the urethral mucosa contain concretions similar to those found within prostatic glands (see Fig. 75.23A). Towards the distal end of the penile urethra, the epithelium changes once again, becoming stratified squamous in type with well-defined connective tissue papillae. This epithelium also lines the navicular fossa and becomes keratinized at the external meatus. The epithelial cells lining the navicular fossa are glycogen-rich. This may provide a substrate for commensal lactobacilli, which, as in the female vagina, provide a defence against pathogenic organisms.

FEMALE URETHRA

The adult female urethra is approximately 4 cm long and 6 mm in diameter. The average length of the urethra is 26 mm in girls, increasing from 23 mm at birth to 32 mm at 15 years (Hirdes et al 2010). It begins at the internal urethral orifice of the bladder, approximately opposite the middle of the pubic symphysis, and runs anteroinferiorly behind the pubic symphysis, embedded in the anterior wall of the vagina (see Fig. 75.1). It is suspended beneath the pubis by the posterior pubourethral ligaments, and anteriorly, by the suspensory ligament of the clitoris. It crosses the perineal membrane and normally ends at the external urethral orifice in the vestibule as an anteroposterior slit with rather prominent margins, directly anterior to the opening of the vagina and 2.5 cm behind the glans clitoris. It sometimes opens into the anterior vaginal wall. Except during the passage of urine, the anterior and posterior walls of the urethra are in apposition and the epithelium is thrown into longitudinal folds, one of which, on the posterior wall of the canal, is termed the urethral crest. Many small, mucous urethral glands and minute, pit-like recesses or lacunae open into the urethra and may give rise to urethral diverticula. On each side, near the lower end of the urethra, a number of these glands, Skene's glands, are grouped together and open into the para-urethral duct; each duct runs down in the submucous tissue and ends in a small aperture on the lateral margin of the external urethral orifice.

Epispadias is a rare congenital anomaly that may affect the female urethra. It may occur in isolation or as part of the exstrophy–epispadias complex. Isolated epispadias in girls is characterized by the urethra opening at the clitoris, which is typically bifid.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Urethral artery

The urethra is supplied principally by the vaginal artery, but also receives a supply from the inferior vesical artery.

Veins

The venous plexus around the urethra drains into the vesical venous plexus around the bladder neck and into the internal pudendal veins (see Fig. 77.3). An erectile plexus of veins along the length of the urethra is continuous with the erectile tissue of the vestibular bulb.

Lymphatic drainage

The urethral lymphatics drain into the internal and external iliac nodes.

INNERVATION

Parasympathetic preganglionic axons arise from neuronal cell bodies in the intermediolateral column of the second to fourth segments of the sacral spinal cord, run in the pelvic splanchnic nerves, and synapse in the vesical plexus in or near the bladder wall (see Fig. 77.5B). Postganglionic fibres are distributed to the smooth muscle of the urethral wall. Somatic fibres to the striated muscle are also derived from the second to fourth segments of the sacral spinal cord, and run in the pelvic splanchnic nerves but do not synapse in the vesical plexus. Sensory fibres run in the pelvic splanchnic nerves to the second to fourth segments of the sacral spinal cord. Postganglionic sympathetic fibres arise from the plexus around the vaginal arteries.

MICROSTRUCTURE

The mucosa lining the female urethra consists of a stratified epithelium and a supporting lamina propria of loose fibroelastic connective tissue. The latter is bulky and well vascularized, and contains numerous thin-walled veins. Its abundant elastic fibres are orientated both longitudinally and circularly around the urethra. The lamina propria contains a fine nerve plexus, believed to be derived from sensory branches of the pudendal nerves. The proximal part of the urethra is lined by urothelium, identical in appearance to that of the bladder neck. Distally, the epithelium changes into a non-keratinizing stratified squamous type that lines the major portion of the female urethra. This epithelium is keratinized at the external urethral meatus, where it becomes continuous with the skin of the vestibule.

The wall of the female urethra consists of an outer muscle coat and an inner mucosa, which lines the lumen and is continuous with that of the bladder. The muscle coat consists of an outer sheath of striated muscle, together with an inner coat of smooth muscle fibres. The female external urethral sphincter is anatomically separate from the adjacent periurethral striated muscle of the anterior pelvic floor. The muscle cells forming the external urethral sphincter are all small-diameter, slow-twitch fibres.

The smooth muscle coat extends throughout the length of the urethra and consists of slender muscle bundles, the majority of which are orientated obliquely or longitudinally. A few circularly arranged muscle fibres occur in the outer aspect of the non-striated muscle layer and intermingle with the skeletal muscle fibres forming the inner part of the external urethral sphincter. Proximally, the urethral smooth muscle extends as far as the bladder neck, where it is replaced by fascicles of detrusor smooth muscle. This region in the female lacks a well-defined circular smooth muscle component comparable with the preprostatic sphincter of the male. Distally, urethral smooth muscle bundles terminate in the subcutaneous adipose tissue surrounding the external urethral meatus.

The smooth muscle of the female urethra receives an extensive presumptive cholinergic parasympathetic nerve supply but contains relatively few noradrenergic nerves. In the absence of an anatomical bladder neck sphincter, competence of the female bladder neck and proximal urethra is unlikely to be totally dependent on smooth muscle activity, and is more probably related to the support provided by the ligamentous structures that surround them. The innervation and longitudinal orientation of most of the muscle fibres suggest that urethral smooth muscle in the female is active during micturition, serving to shorten and widen the urethral lumen.

MICTURITION AND URINARY CONTINENCE

The central integration of nervous control of the bladder and urethra is essential for normal micturition (Fig. 75.16). (For further reading, see Yoshimura and Chancellor (2012).)

Micturition consists of storage and voiding phases. During the storage phase, the bladder accommodates an increasing volume of urine

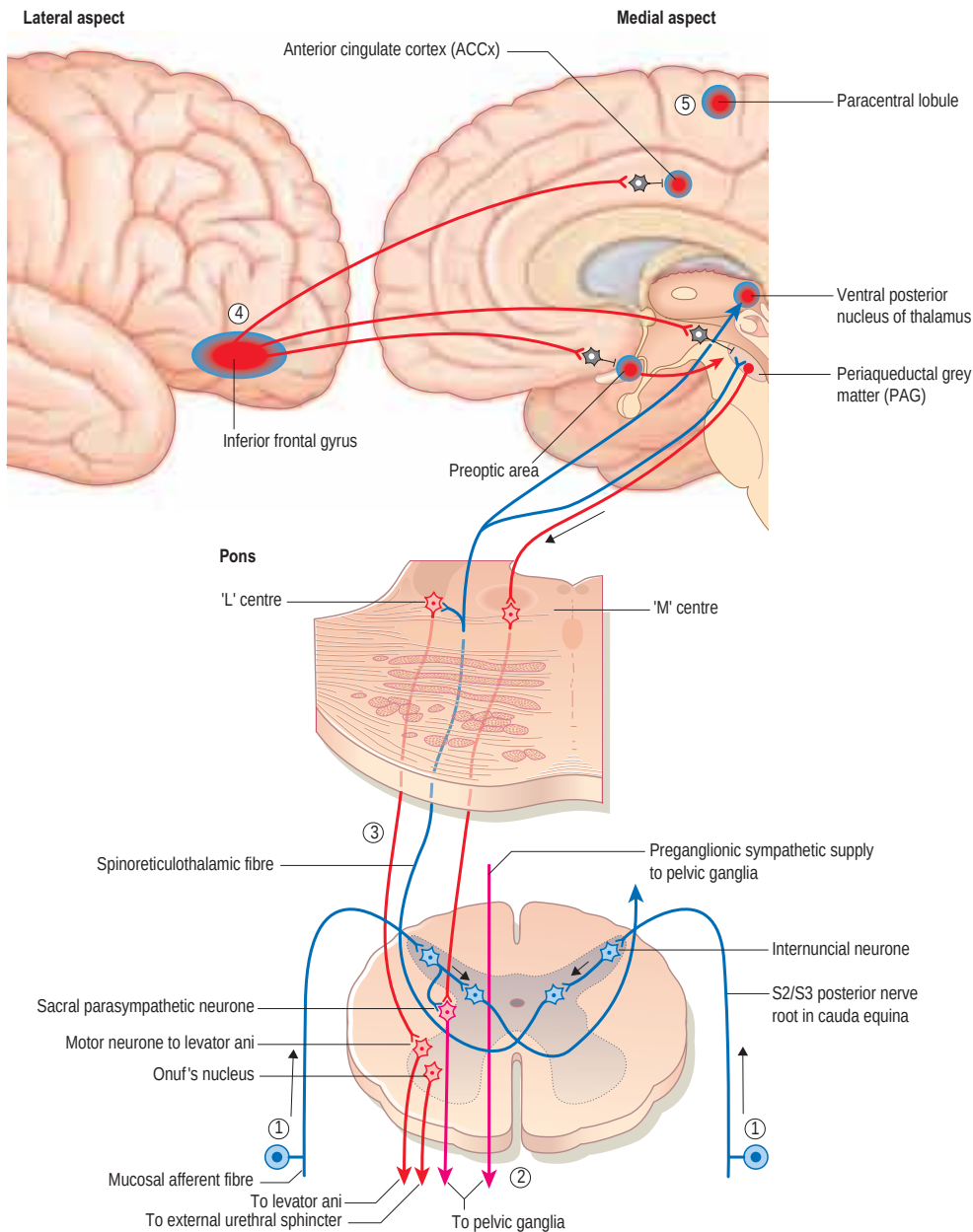


Fig. 75.16 The micturition control centre is in the paramedian pontine reticular formation on each side and consists of a medially placed micturition centre, 'M', and a laterally placed storage centre, 'L'. Neurones project from the 'M' centre and the storage centre 'L' to parasympathetic neurones in segments 2–4 of the sacral spinal cord, and to Onuf's nucleus, which is in the same segments, and which innervates the external urethral sphincter. At higher levels, neurones in the right prefrontal and anterior cingulate cortex, right preoptic nucleus and periaqueductal grey matter are involved in the control of micturition. Vesical afferents from stretch receptors in the detrusor and trigonal mucosa relay the extent of bladder filling to the brainstem and thalamus via spinoreticulothalamic fibres (1). Activity in the sympathetic system that maintains increases in bladder compliance (via β_2 receptors on detrusor fibres) and parasympathetic activity is inhibited (2). Spinoreticular fibres synapsing in the 'L' nucleus in the pons activate Onuf's nucleus to increase the tone of the external sphincter (3). If micturition is deferred, fibres projecting from the inferior frontal gyrus inhibit the right anterior cingulate gyrus, preoptic area and periaqueductal grey matter (4). Voluntary contraction of the pelvic floor musculature, controlled by the prefrontal cortex driving the perineal 'area' of the motor cortex (5), cannot be long sustained once filling is complete. (With permission from FitzGerald MJT, Folan-Curren J 2001 *Clinical Neuroanatomy*, 4th edn. London: Saunders.)

without any change in intravesical pressure, partly because of the viscoelastic properties of its walls, and partly because a gating mechanism operates in the spinal cord that reflexively inhibits preganglionic parasympathetic activity. A gating mechanism in the pelvic ganglia prevents the activation of postganglionic parasympathetic neurones until preganglionic activity has reached a threshold level (Chancellor and Yoshimura 2002).

Mean bladder capacity in adult males varies around 400 ml but micturition commonly occurs at smaller volumes. Voluntary control is imposed from the inferior frontal gyrus of the cerebral cortex. Filling to 500 ml may be tolerated; beyond this level, pain caused by tension in the bladder wall leads to the urgent desire to micturate. The pain is referred to the cutaneous areas supplied by T10–L2 and S2–4, including the lower anterior abdominal wall, perineum and penis. Threshold afferent stimulation activates the micturition centre in the rostral pons (the 'M' centre) (see Fig. 75.16), which drives preganglionic parasympathetic neurones in the intermediolateral grey column of the second, third and fourth sacral spinal segments via descending spinal pathways. The axons of these neurones run to the inferior hypogastric plexus in the pelvic splanchnic nerves; they synapse on postganglionic neurones in ganglia lying within the plexus and in the wall of the bladder. Postganglionic axons ramify throughout the thickness of the detrusor smooth muscle coat. When stimulated, they release acetylcholine, which activates muscarinic receptors in the detrusor layer of the bladder wall and produces the sustained bladder contraction required for micturition. The distal urethral sphincter maintains urethral closure.

Urinary continence in the male

Urinary continence at the level of the membranous urethra is mediated by the radial folds of urethral mucosa, the submucosal connective tissue, the intrinsic urethral smooth muscle, the striated external urethral sphincter and the pubourethral component of levator ani, puboperinealis (Fig. 75.17). The external urethral sphincter represents the point of highest intraurethral pressure in the normal, contracted, state. The striated muscle component of the external urethral sphincter is devoid of muscle spindles. The striated muscle fibres themselves are unusually small in cross-section (15–20 μ m in diameter), and are physiologically of the slow-twitch type, unlike the pelvic floor musculature, which is a heterogeneous mixture of slow- and fast-twitch fibres of larger diameter. The slow-twitch fibres of the external sphincter are capable of sustained contraction over relatively long periods of time and actively contribute to the tone that closes the urethra and maintains urinary continence. They are innervated by neurones that lie in Onuf's nucleus in the anterolateral grey matter of the second to fourth sacral spinal segments. Their firing is controlled centrally by a storage centre within the rostral pons (the 'L' centre) (see Fig. 75.16). Just before the onset of voiding, the external urethral sphincter is relaxed by central inhibition of Onuf's nucleus.

Urinary continence in the female

The urethral sphincter mechanism consists of the intrinsic striated and smooth muscle of the urethra, the mucosa and submucosal connective tissue, and the puborectalis component of levator ani (which surrounds the urethra at the point of maximum concentration of those muscles).



Fig. 75.17 **A**, Coronal illustration of male anterior levator ani (urogenital) hiatus: prostate above, intervening striated urethral sphincter and perineal membrane with corporal bodies below. **B**, This coronal T2-weighted MRI includes the cephalad extension of the smooth muscle urethral sphincter, which cannot be discriminated from striated urethral sphincter; bulbourethral glands are offset from expected symmetry. (**A**, **B** Courtesy of Robert P Myers MD, Akira Kawashima MD and Bernard F King MD, used with permission of Mayo Foundation for Medical Education and Research.)

The striated muscle component of the urethral sphincter mechanism surrounds the middle and lower thirds of the urethra in the female. Proximally, it forms a complete ring around the urethra, while, more distally, it covers the anterior and lateral aspects of the urethra; it blends above with the smooth muscle of the bladder neck and below with the smooth muscle of the lower urethra and vagina. Contraction of this part of the sphincter compresses the urethra against the relatively fixed anterior vaginal wall. At its most distal point; the striated sphincter encompasses the urethra and vagina as the urethrovaginal sphincter. The mucosa and submucosa are oestrogen-dependent and atrophy postmenopausally, possibly resulting in stress incontinence.

PROSTATE

The prostate is a globular fibromuscular gland that surrounds the prostatic urethra from the bladder base to the membranous urethra (Myers 2001). It is enclosed by firmly adherent tissue that has been variously termed the prostatic capsule or the prostatic fascia (parietal fascia covering the adjacent part of levator ani, lateral pelvic fascia, periprostatic fascia, parapelvic fascia), reflecting, in part at least, inter-individual variability and sampling site (Myers et al 2010) (Fig. 75.18). Histologically, it is a multilayered connective tissue (Hinata et al 2013). There is no true fibrous capsule at the base or apex of the prostate, and the presence of a capsule between the apex and base is variable.

The muscular tissue within the prostate is mainly smooth muscle. Anterior to the urethra, a layer of smooth muscle merges with the main mass of muscle in the fibromuscular septa; it blends superiorly with vesical smooth muscle. Anterior to the layer of smooth muscle, a transversely crescent-shaped mass of skeletal muscle is continuous inferiorly with the external urethral sphincter in the deep perineal pouch. Its fibres pass transversely internal to the capsule and are attached to it laterally by diffuse collagen bundles; other collagen bundles pass posteromedially, merging with the prostatic fibromuscular septa and the septum of the urethral crest. This muscle, supplied by the pudendal nerve, probably compresses the urethra but it may pull the urethral crest back and the prostatic sinuses forwards, dilating the urethra. Glandular contents may be expelled simultaneously into the urethra when it has expanded in this way, so that it contains 3–5 ml seminal fluid prior to ejaculation.

The prostate lies at a low level in the lesser pelvis, behind the inferior border of the pubic symphysis and pubic arch (see Fig. 75.2A; Fig. 75.19), and anterior to rectourethralis and the rectal ampulla, through which it may be palpated. It presents a base or vesical aspect superiorly, an apex inferiorly, and posterior, anterior and two inferolateral surfaces. The prostatic base measures about 4 cm transversely. The gland is 2 cm in its anteroposterior, and 3 cm in its vertical, diameters. It weighs

about 8 g in youth but, almost invariably, enlarges with the development of BPH; it usually weighs 40 g, but sometimes as much as 150 g or even more, after the first five decades of life. The small prostate without BPH is described as a croissant shape (short anterior commissure, prominent apical notch and posterior lip of prostatic tissue), and the enlarged gland is more doughnut-shaped. The shape of the prostate affects the relationship of the prostatic apex to the external urethral sphincter. This relationship is important when removing the prostate at radical prostatectomy for cancer, and anastomosing the bladder to the urethra to maintain sphincter integrity. The external urethral sphincter is flush to a large doughnut-type gland, and so a perpendicular incision will separate the prostate and external urethral sphincter accurately. In a small prostate, the external urethral sphincter fills the defect in the anterior aspect of the prostate, and so a perpendicular incision at the level of the posterior lip of the croissant-shaped gland will excise much external urethral sphincter and leave the patient incontinent.

Superiorly, the base is largely contiguous with the neck of the bladder. The apex is inferior, surrounding the junction of the prostatic and membranous parts of the posterior urethra. The apical posterior relation of the prostate and external urethral sphincter is rectourethralis, a Y-shaped muscle that originates from the outer longitudinal coat of the rectum. The upper limbs of the Y extend from the lateral rectal wall to the midline, where rectourethralis inserts into the perineal body at the anorectal junction (Brooks et al 2002).

The anterior surface lies in the arch of the pubis, separated from it by the dorsal vascular complex (Santorini's plexus) and loosely attached adipose tissue. It is transversely narrow and convex, extending from the apex to the base. Near its superior limit, it is connected to the pubic bones by the puboprostatic ligaments. The urethra emerges from this surface anterosuperior to the apex of the gland. The anterior part of the prostate is relatively deficient in glandular tissue and is largely composed of fibromuscular tissue. The anterior and lateral aspects of the prostate are covered by a layer of fascia derived from the endopelvic fascia on each side, called the lateral prostatic fascia. This is adherent medially to the prostate, continues posteriorly over the lateral aspect of the prostate, neurovascular bundles and rectum (lateral rectal fascia), and passes distally over the urethra (see Fig. 75.18A). The prostatic venous plexus (see Fig. 75.19) lies between this extension of the endopelvic fascia and the prostate. Anteroinferiorly, the parietal and visceral fasciae of the prostate merge and blend with the puboprostatic ligaments. The anterior surface of the prostate and associated vascular plexus is covered by the detrusor apron (Myers 2002).

The inferolateral surfaces are related to the muscles of the pelvic side wall; the anterior fibres of levator ani embrace the prostate in the pubourethral sling or pubourethralis. These muscles are separated from the prostate by a thin layer of connective tissue.

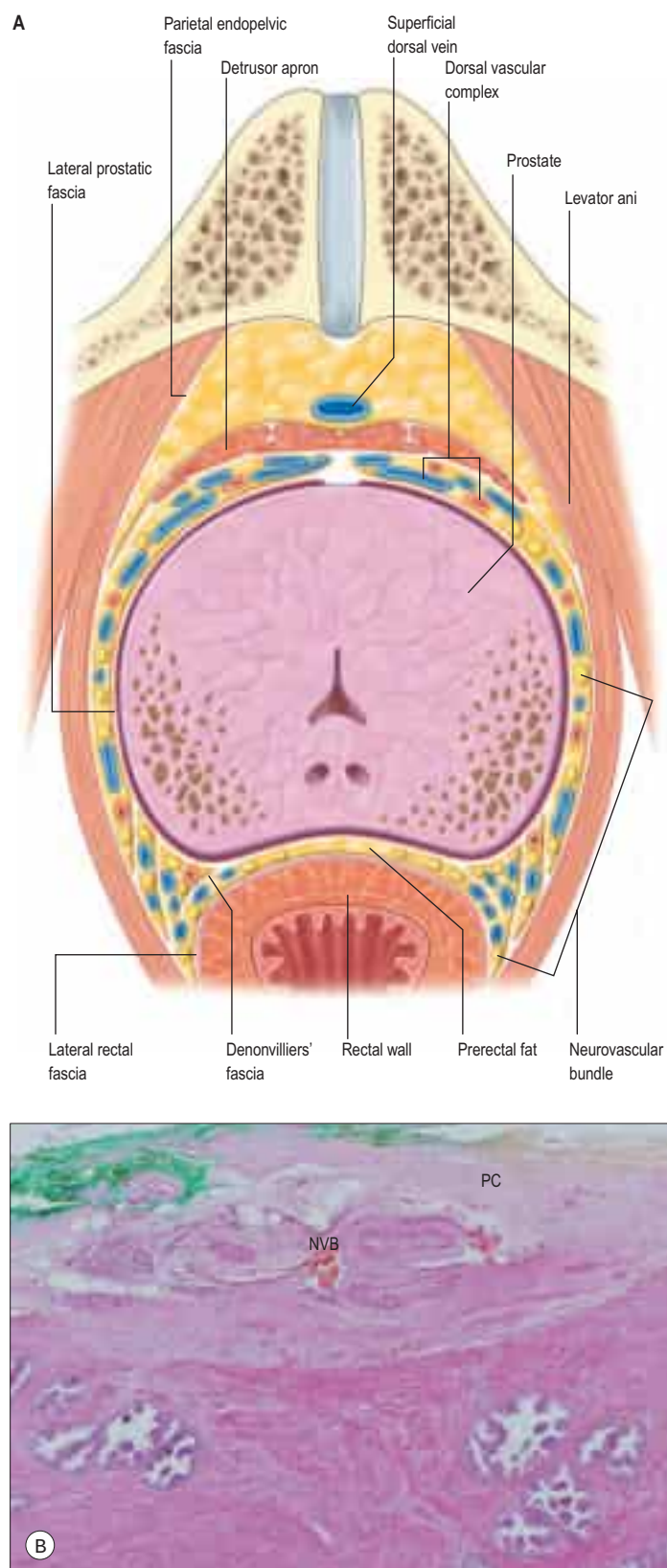


Fig. 75.18 A, The endopelvic fascia. The fascia covers the pelvic viscera and continues posteriorly over the lateral aspect of the prostate, as the lateral prostatic fascia and, on the lateral aspect of the rectum, as the lateral rectal fascia. Its relationship with Denonvilliers' fascia may be visualized as an 'H'. The deep dorsal vein and lateral branches run deep to the endopelvic fascia and the lateral prostatic fascia, although communicating with perforators to the pelvic side wall. Denonvilliers' fascia is adherent to the prostate in the posterior midline but, like the lateral prostatic fascia, is separated from the prostatic tissue by the neurovascular structures and fatty tissues elsewhere. Between Denonvilliers' fascia and the rectum is a fatty potential plane, the prerectal space. **B**, The prostatic capsule (PC) is a condensation of connective tissue (green) on the surface of the gland and encloses the neurovascular bundle (NVB). The prostatic fibromuscular stroma and glands lie beneath the capsule.

The posterior surface of the prostate is transversely flat and vertically convex. Near its superior (juxtavesical) border is a depression where it is penetrated by the two ejaculatory ducts. Below this is a shallow, median sulcus, usually considered to mark a partial separation into right and left lateral lobes. It is separated from prerectal fat in the prerectal space (see Figs 75.2A, 75.18A) and rectum by Denonvilliers' fascia, a condensation of pelvic fascia that develops by obliteration of the rectovesical peritoneal pouch, and by loose yellow, fatty areolar tissue. The rectovesical pouch is obliterated from below upwards as fetal life progresses, forming Denonvilliers' fascia; at birth, this fascia separates the prostate, the seminal vesicles and the ampullae of the vasa deferentia from the rectum. The superior limit of Denonvilliers' fascia is the peritoneum of the rectovesical pouch. Laterally, Denonvilliers' fascia fuses with the lateral pelvic fascia; anterior to Denonvilliers' fascia, the lateral pelvic fascia is called the lateral prostatic fascia and, posterior to Denonvilliers' fascia, it is called the lateral rectal fascia.

The prostate is traversed by the urethra and ejaculatory ducts, and contains the prostatic utricle. The urethra enters the prostate near its anterior border and usually passes between its anterior and middle thirds. The ejaculatory ducts pass anteroinferiorly through its posterior region to open into the prostatic urethra.

ZONAL ANATOMY OF THE PROSTATE



The prostate gland was once thought to be divided into five anatomical lobes, but it is now recognized that five lobes can only be distinguished in the fetal gland prior to 20 weeks' gestation. Between then and the onset of BPH, only three lobes are recognizable: two lateral and a median lobe. Clinicians refer to left and right 'lobes' when describing either what can be felt on rectal palpation, or the endoscopically visible abnormalities that are seen in the diseased state when prostatic anatomy is distorted by BPH.

From an anatomical, and particularly from a morbid anatomical, perspective, the glandular tissue may be subdivided into three distinct zones (Figs 75.20–75.22): peripheral (70% by volume), central (25% by volume) and transitional (5% by volume) (Mundy et al 1999b). Non-glandular tissue (fibromuscular stroma) fills the space between the peripheral zones anterior to the preprostatic urethra. The central zone surrounds the ejaculatory ducts, posterior to the preprostatic urethra, and is more or less conical in shape, with its apex at the verumontanum. The fibromuscular stroma includes the smooth muscle detrusor apron superoanteriorly and striated urethral sphincter fibres that extend anteriorly over the anterior prostate on to the bladder neck. The transitional zone lies around the distal part of the preprostatic urethra just proximal to the apex of the central zone and the ejaculatory ducts; it includes the smooth muscle of the preprostatic urethral wall. Its ducts enter the prostatic urethra just below the preprostatic sphincter and just above the ducts of the peripheral zone. The peripheral zone is cup-shaped and encloses the central transitional zone and the preprostatic urethra, except anteriorly, where the space is filled by the anterior fibromuscular stroma. Simple mucus-secreting glands lie in the tissue around the preprostatic urethra, above the transitional zone and surrounded by the preprostatic sphincter. These simple glands are similar to those in the female urethra and unlike the glands of the prostate.

On magnetic resonance imaging (MRI), the prostate gland has a zonal anatomy on T2-weighted images (see Fig. 75.21). The normal peripheral zone has high-signal intensity, as does fluid within the seminal vesicles. The central and transitional zones have relatively low signal and are often referred to as the 'central gland'. The verumontanum may be seen as high-signal within the central gland. The relationship of the zones of the gland normally changes with age. The central zone atrophies, and the transitional zone enlarges secondary to BPH. This often produces a low-signal band at the margin of the hypertrophied transitional and compressed peripheral zones: the surgical pseudocapsule, which is well seen on T2-weighted MR images.

The zonal anatomy of the prostate is clinically important because most carcinomas arise in the peripheral zone, whereas BPH affects the transitional zone, which may grow to form the bulk of the prostate.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The prostate is supplied by branches from the inferior vesical, internal pudendal and middle rectal arteries (see Fig. 75.19). They perforate the gland along a posterolateral line from the junction of the prostate with the bladder down to the apex of the gland.

As the transitional zone grows, it produces the appearance of 'lobes' on either side of the urethra. In due course, these lobes may compress or distort the preprostatic and prostatic parts of the urethra and produce symptoms. The central zone surrounding the ejaculatory ducts is rarely involved in any disease. It shows certain histochemical characteristics that differentiate it from the rest of the prostate; it is thought to be derived from the Wolffian duct system (much like the epididymi, vasa deferentia and seminal vesicles), whereas the rest of the prostate is derived from the urogenital sinus ([p. 1218](#)).

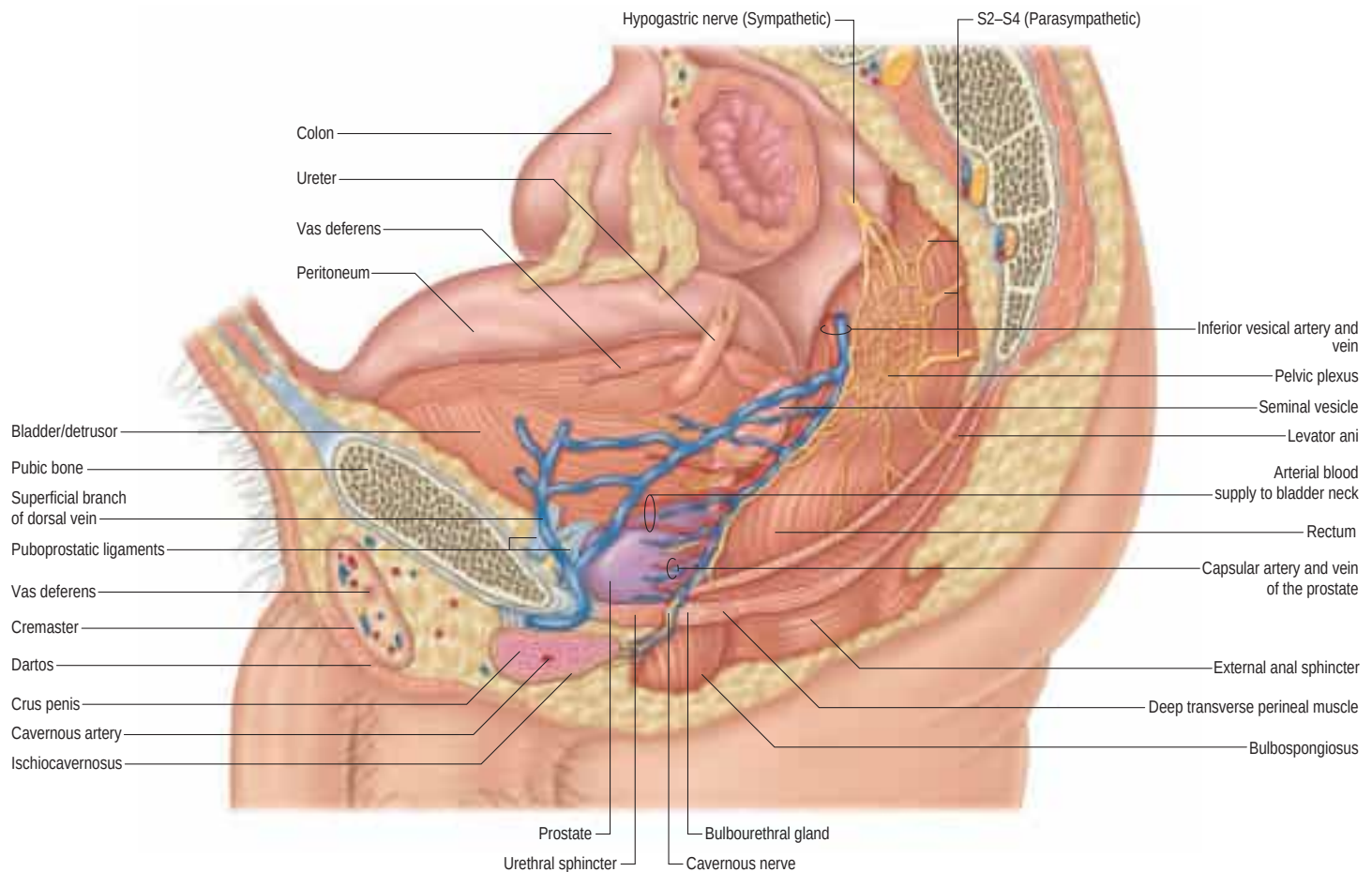


Fig. 75.19 A sagittal section of the male pelvis. The pelvic plexus lies with its midpoint level with the tip of the seminal vesicles. It gives branches to the prostate, bladder urethra, seminal vesicles, rectum and corpora cavernosa via the cavernous nerves. The cavernous nerves are the continuation of the neurovascular bundles, lying posterolateral to the prostate. These bundles may be damaged at radical prostatectomy, resulting in impotence.

The inferior vesical artery often arises from the internal iliac artery with the middle rectal artery. It gives rise to two groups of branches: the urethral and capsular. The urethral vessels enter at the prostatovesical junction, principally posteriorly at the 5 and 7 o'clock positions, but also anteriorly at 1 and 11 o'clock. This bladder neck arterial anatomy is always apparent at transurethral resection of the prostate and open removal of BPH adenomas. The capsular arteries run posterolaterally and inferiorly in the neurovascular bundles, providing perpendicular perforating vessels to the prostate. The most constant in position and prominence is the apical perforator at the prostatourethral junction, an important landmark for this point and for the neurovascular bundle at radical prostatectomy.

Veins

The veins run into a plexus around the anterolateral aspects of the prostate, posterior to the arcuate pubic ligament and the lower part of pubic symphysis, anterior to the bladder and prostate. The chief tributary is the deep dorsal vein of the penis. The plexus also receives anterior vesical and prostatic rami (which connect with the vesical plexus and internal pudendal vein), and drains into vesical and internal iliac veins.

Lymphatic drainage

Collecting vessels from the vas deferens drain into the external iliac nodes, while those from the seminal vesicle drain to the internal and external iliac nodes. Prostatic vessels end mainly in internal iliac, sacral and obturator nodes. A vessel from the posterior surface accompanies the vesical vessels to the external iliac nodes, and another from the anterior surface reaches the internal iliac group by joining vessels that drain the membranous urethra.

INNERVATION

The prostate receives an abundant nerve supply from the inferior hypogastric (pelvic) plexus (see Figs 59.4, 75.11 and Fig. 8.1.1 in

Commentary 8.1). The prostatic capsule is covered by numerous nerve fibres and ganglia posterolaterally, forming a crescentic periprostatic nerve plexus. The greatest density of nerves is found in the preprostatic sphincter; fewer fibres are found in the anterior fibromuscular stroma, and the peripheral zone is the least densely innervated. Nerves containing neuropeptide Y and vasointestinal polypeptide (VIP) are localized in the subepithelial connective tissue, in the smooth muscle layers of the gland, and in the walls of its blood vessels. Neurovascular bundles (Walsh and Donker 1982) containing autonomic nerves that supply the prostate, seminal vesicles, prostatic urethra, ejaculatory ducts, corpora cavernosa, corpus spongiosum, membranous and penile urethra, and bulbourethral glands are closely applied to, but separable from, the posterolateral margins of the prostate (see Commentary 8.1). They are intimately related to the prostatic fascia. These nerves may be damaged during radical prostate surgery for organ-confined prostate cancer, producing impotence, or sacrificed as part of wide local excision of the prostate (Tewari et al 2006, Pisipati et al 2014).

The somatic pudendal nerve supplies the external urethral sphincter. The branches enter at the 5 and 7 o'clock positions. Sensory branches pass through the penile hilus in association with the dorsal venous complex and dorsal vein of the penis.

MICROSTRUCTURE

The glandular tissue consists of numerous follicles with frequent internal papillae. Follicles open into elongated canals, which join to form 12–20 main ducts. The follicles are separated by loose connective tissue, supported by extensions of the fibrous capsule and muscular stroma, and enclosed in a delicate capillary plexus. Follicular epithelium is variable but predominantly columnar, and either single-layered or pseudostratified.

Prostatic ducts open mainly into the prostatic sinuses in the floor of the prostatic urethra. They have a bilayered epithelium, the luminal layer is columnar, and the basal layer is populated by small cuboidal cells. Small colloid amyloid bodies (corpora amylacea) are frequent in the follicles (Fig. 75.23). Prostatic and seminal vesicular

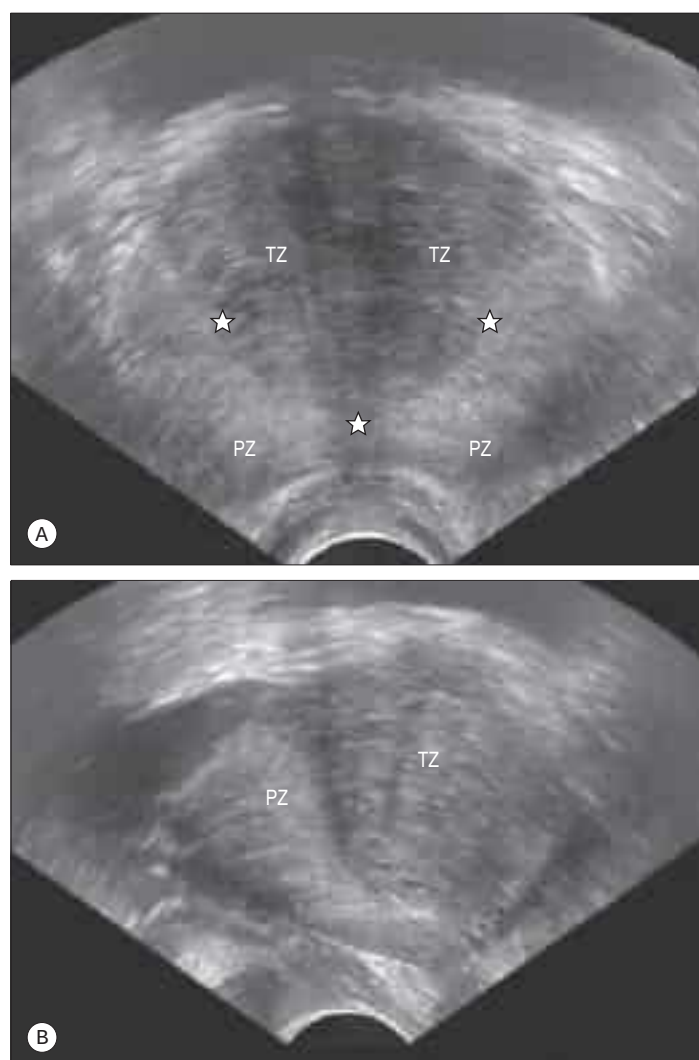


Fig. 75.20 Ultrasound of the prostate. **A**, An axial view showing the hypoechoic transition zone (TZ) and the more echogenic peripheral zone (PZ). Their interface is the surgical capsule (*). **B**, A sagittal view showing the hypoechoic transition zone (TZ) and the more echogenic peripheral zone (PZ).

secretions form the bulk of seminal fluid. Prostatic secretions are slightly acid, and contain acid phosphatase, amylase, prostate-specific antigen, fibrinolysin and zinc. Numerous neuroendocrine cells, containing neurone-specific enolase, chromogranin and serotonin (5-hydroxytryptamine, 5-HT), are present in the glandular epithelium; their numbers decline after middle age and their function is unknown.

Histological sections just above the level of the verumontanum reveal two concentric, partially circumferential, zones of glandular tissue. The larger outer zone is the peripheral zone and has long, branched glands, whose ducts open mainly into the prostatic sinuses. The inner zone is the transitional zone and consists of glands whose ducts open on the floor of the prostatic sinuses and colliculus seminalis, and a group of simple mucosal glands that surround the preprostatic urethra. Anteriorly, in the prostatic isthmus, the peripheral zone and submucosal glands are absent.

AGE CHANGES IN THE PROSTATE

At birth, the prostate has a system of ducts embedded in a stroma that forms a large part of the gland. Follicles are represented by small end-buds on the ducts. Before birth, the epithelium of the ducts, seminal colliculus and prostatic utricle display hyperplasia and squamous metaplasia, possibly due to maternal oestrogens in the fetal blood. This subsides after birth and is followed by a period of quiescence lasting for 12–14 years.

At puberty, between the ages of approximately 14 and 18 years, the prostate gland enters a maturation phase and more than doubles in size during this time. Growth is almost entirely due to follicular development, partly from end-buds on ducts, and partly from modification of

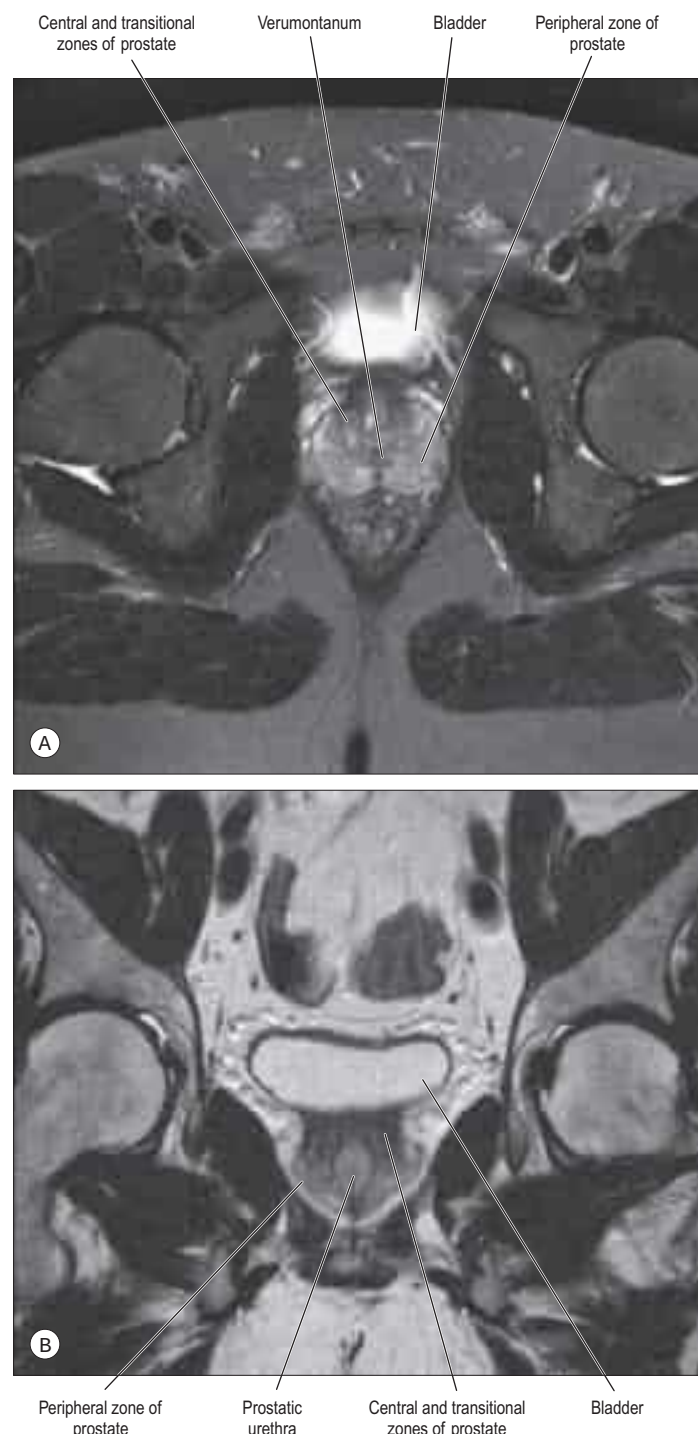


Fig. 75.21 Magnetic resonance imaging (MRI) of the prostate. **A**, A T2-weighted MRI scan showing the normal high signal of the peripheral zone and the intermediate signal of the central and transitional zones and the verumontanum in the central gland. **B**, A T2-weighted coronal MRI scan of the prostate showing the zonal anatomy.

the ductal branches. Morphogenesis and differentiation of the epithelial cords starts in an intermediate part of the epithelial anlage and proceeds to the urethral and subcapsular parts of the gland; the latter is reached by the age of 17–18 years. The glandular epithelium is initially multi-layered squamous or cuboidal, and is transformed into a pseudostratified epithelium consisting of basal, exocrine secretory (including mucous) and neuroendocrine cells. The mucous cells are temporary, and are lost as the gland matures. The remaining exocrine secretory cells produce a number of products, including acid phosphatase, prostate-specific antigen and β -microseminoprotein. Growth of the secretory component is associated with condensation of the stroma, which diminishes relative to the glandular tissue. These changes are probably a response to the secretion of testosterone by the testis.

During the third decade, the glandular epithelium grows by irregular multiplication of the epithelial infoldings into the lumen of the follicles. After the third decade, the size of the prostate remains virtually

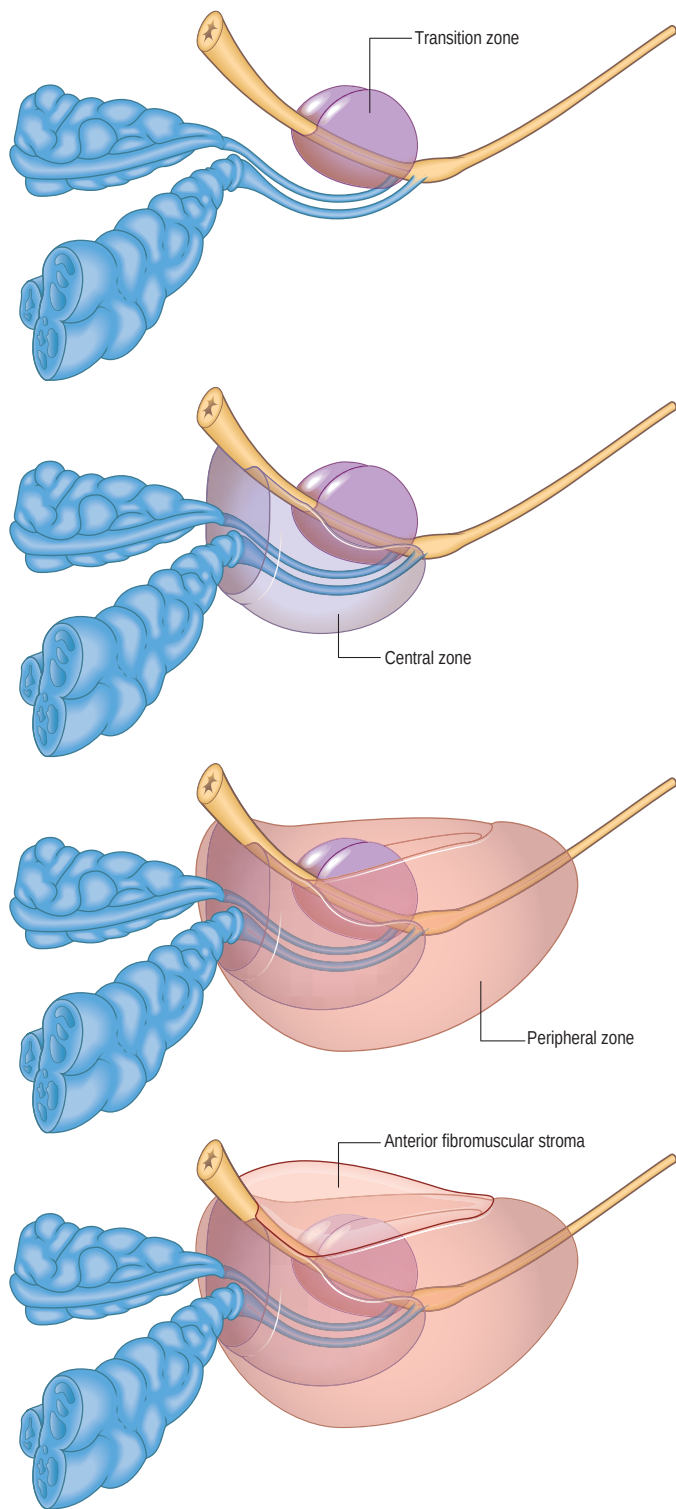


Fig. 75.22 The zonal anatomy of the prostate. (With permission from Walsh PC, Retik AB, Vaughan ED et al (eds) 2002 Campbell's Urology, 8th edition. Philadelphia: Saunders.)

unaltered until 45–50 years, when the epithelial foldings tend to disappear, follicular outlines become more regular, and amyloid bodies increase in number: all signs of prostatic involution. After 45–50 years, the prostate tends to develop BPH: an age-related condition. If a man lives long enough, then BPH is inevitable, although not always symptomatic.

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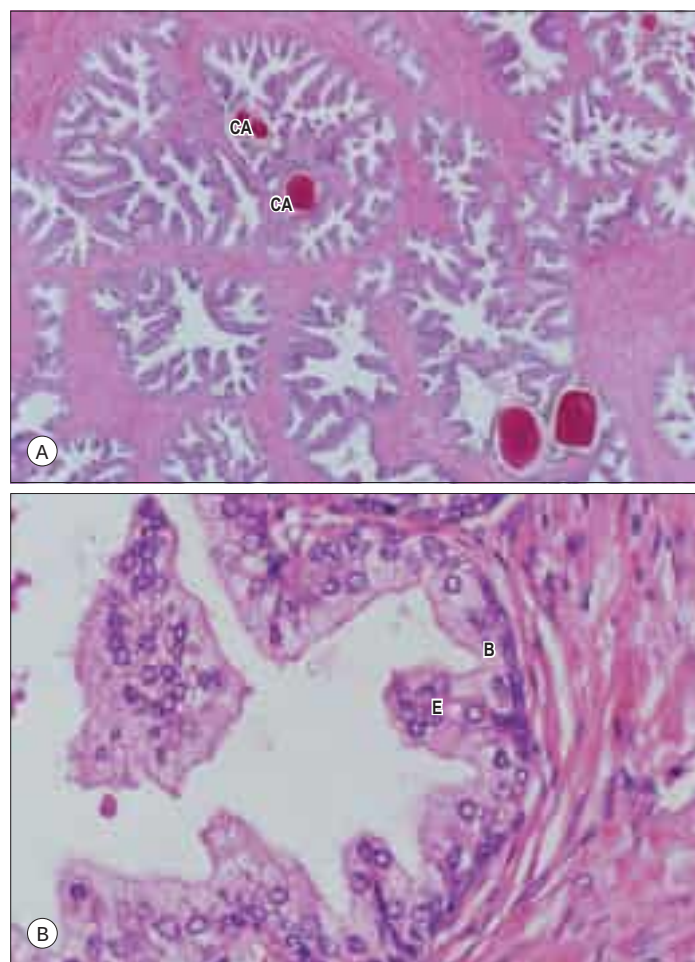


Fig. 75.23 **A**, Prostatic acini show papillary infoldings at the base of the gland (central zone). They are irregularly divided by fibromuscular stroma. Some acini contain eosinophilic (pink) secretions called corpora amylacea (CA) ($\times 10$, haematoxylin and eosin stain). **B**, Prostatic acini consist of a double layer of epithelial cells (E), which line the lumen, and basal cells (B), which give rise to epithelial cells.



Bonus e-book images and videos

Fig. 75.6 Centiles of bladder volume index in children.

Fig. 75.8 A laparoscopic view of the empty bladder and median and medial umbilical folds in a 12-month-old boy.

Fig. 75.9 A laparoscopic view of the left inferior epigastric vein (lateral umbilical ligament) in a 12-month-old boy.

Fig. 75.14 Cystic dilation of the prostatic utricle demonstrated in an 8-month-old child during a micturating cystourethrogram.

Video 75.1 Laparoscopic view of bladder filling and emptying in relation to the rectovesical pouch.

Video 75.2 Laparoscopic view of anterior abdominal wall and ligaments.

BPH begins as micronodules in the transitional zone, which grow and coalesce to form macronodules around the inferior margin of the preprostatic urethra, just above the verumontanum. Macronodules, in turn, compress the surrounding normal tissue of the peripheral zone posteroinferiorly, creating a 'false capsule' around the hyperplastic tissue, which, coincidentally, provides a plane of cleavage for its surgical enucleation (for further reading see [Kutikov et al \(2006\)](#)).

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